

Near-miss constructions, interaction surfaces, and prime-support obstructions for the 3×3 magic square of squares

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Abstract

We study near misses to a 3×3 magic square of nine distinct positive integer squares through the common difference of its transversal progressions. Using the complete S -unit totals of von Känel and Matschke, we reconstruct the corresponding square progressions and exclude fully magic squares whose transversal difference has only prime factors at most 19. In the same domain we classify exactly 384 primitive nine-square arrays with seven equal line sums, up to grid symmetry and common square scaling, and determine their sharp difference-normalized defect. A separate consequence of Aebi’s triangle classification gives uniqueness of a positive square progression for every difference with at most three prime factors, without a primitivity assumption. For arrays with eight square entries, we construct two infinite families: a retraction of classical six-square carriers with a complete least-shift law, and an elliptic-orbit lift with an explicit positive index density. We also classify a finite bridge of difference 840. For the eighth-line condition we distinguish a fixed-difference elliptic interaction from a variable-difference model modulo scale; the latter is an irreducible surface whose quintic reconstruction over the square-progression cone has generic Galois group S_5 . We specialize the finiteness theorem of Caro and García-Fritz, close particular arithmetic fibres, and derive local support restrictions. The general existence problem and arbitrary four-prime supports remain open. Exact verifiers and identified external computer-algebra records accompany the proofs.

1 Introduction

Does there exist a 3×3 magic square whose nine entries are distinct positive integer squares? LaBar posed this question in 1984; Gardner popularised it and offered \$100 for an example (see the historical account in Bremner [Bre99], §0). Two classical near misses approach the question in different ways: Bremner’s fully magic array has seven square entries [Bre99], whereas Sallows’s nine-square array has seven equal ordinary line sums [Sal97]. Here an ordinary line means a row, column, or long diagonal. We call an array an $(8, 7)$ *array* if its nine entries are positive and pairwise distinct, exactly eight are integer squares, and exactly seven ordinary lines have one common sum. A $(9, 7)$ *array* satisfies the same conditions with all nine entries square.

The common structure is elementary but arithmetically useful: an array with seven equal line sums consists of three arithmetic progressions with one common difference. Closing its eighth line asks that their three centres form another arithmetic progression. This separates two questions: how to construct near misses, and which differences can support a fully magic interaction. The prime support in this paper belongs to the *difference*, not to the nine entries.

1.1 Contributions and their dependencies

Complete fixed-support results. Our most concrete classification is Theorem 3. Using the complete S -unit totals of von Känel–Matschke [vKM16], we give a bidirectional reconstruction of

the relevant primitive right triangles and square progressions, test every possible centre interaction, and classify the resulting $(9, 7)$ arrays. For primes at most 19 this gives no fully magic interaction, exactly 384 classes modulo the dihedral group and common rational-square scaling, and the unique minimum

$$\eta = \frac{|\text{common line sum} - \text{failed line sum}|}{|\text{transversal difference}|} = \frac{347984603}{9837828000}.$$

The imported total certifies completeness beyond the finite production bound. The reconstruction, classification and optimization are the finite deductions proved here. In particular, the small-support example already displayed by Houston [Hou16b] is credited as such; the atlas classifies the entire stated domain. Separately, the seven triangles in Aebi’s classification [Aeb26] have distinct area squareclasses. We deduce that a difference with at most three prime factors supports at most one positive square progression (Theorem 2).

Constructive results. We determine the integral image lattice of the line-sum map and prove its Smith form, refining the classical normal form (Proposition 1). Starting with catalogued Bremner–Wesolowski carriers, we then prove a uniform retraction formula, classify every pair-closing shift and determine the least one. A second infinite construction lifts an elliptic orbit to pairwise inequivalent primitive $(8, 7)$ arrays; applying Cowan’s equidistribution theorem [Cow20] gives the density of the admissible *orbit indices*. A finite classification of the difference-840 bridge supplies a further explicit construction. These results form Theorem 4: two infinite mechanisms and one finite bridge, with the classical carriers and elliptic ingredients attributed at use.

The interaction and its arithmetic restrictions. We develop the centre-interaction condition in two settings. For fixed difference 840 it is a relation on three doubled elliptic points. Allowing the difference to vary and dividing out the common root scale gives a different, explicitly related surface. We prove that this normalized surface is irreducible and that its degree-five reconstruction over the square-progression cone has generic Galois group S_5 (Theorem 5). For a prescribed difference an additional rational-square condition on the scale is essential. We specialize Caro–García–Fritz’s theorem [CGF25] to obtain non-effective fixed-squareclass finiteness, close the centre-841 fibre and a fixed Bremner shadow by explicit elliptic quotients, and give a countercertificate to deciding the ambient interaction from the three separate finite Kummer classes alone. Finally we derive a nondegenerate five-term S -unit equation, local valuation restrictions and precise reductions of the remaining four-prime branches. The finite-support catalogue above is a separate effective argument; it does not depend on the non-effective finiteness bound.

1.2 Relation to earlier work and organization

The normal form and its interpretation by square progressions occur in Houston [Hou16a, Hou16b], following Lucas [Luc76, Luc91]; the equal-area triangle correspondence is classical and is used by Robertson [Rob96]. Bremner [Bre99, Bre01], Boyer’s survey and catalogue [Boy05, Boy], and Fituvalu’s tables [Fit] supply the carrier context. The support law uses the unit-equation theorem of Evertse–Schlickewei–Schmidt [ESS02]. These inputs are distinguished from the refinements, constructions and applications proved in this paper. Results restricting the prime factors of the *entries*, due to Rabern, Woll, Labruna and Weisenberg [Rab03, Wol18, Lab18, Wei23], concern a complementary quantity; Appendix D records their scope and further bibliographic comparisons. It also explains the elementary quartic-field completion of Bremner’s example, which shows why a shape-only obstruction cannot settle the rational problem. The historical label “Parker” in some

companion records recalls Parker’s popular near miss [Par]; it neither attributes LaBar’s problem to him nor identifies his particular array with the configurations studied here.

Section 2 fixes the normal form and conventions. Section 3 proves the support exclusions and exact atlas, followed by the constructions in Section 4. Sections 5 and 6 treat the interaction geometry and specific arithmetic obstructions. Sections 7 and 8 develop the local support laws and four-prime frontier; Section 9 identifies the remaining questions. The appendices retain the bounded census, least-shift details, supplementary reductions, source comparisons and reproducibility information. External rank and point-list computations are identified where they enter a proof; their archived output records are distinct from the exact computations rerun by the public verifiers.

2 The seven-line form and square progressions

We write D_4 for the eight rotations and reflections of the grid. Two positive rational arrays are square-scale equivalent if one is a common nonzero rational square multiple of the other, possibly followed by an element of D_4 . An all-squares array has a unique positive integral-root normalization with root gcd 1 before taking the grid quotient. For an $(8, 7)$ array, *root-primitive* refers to the gcd of its eight square roots, and *entry-primitive* to the gcd of all nine integral entries.

Houston [Hou16a, Hou16b], starting from the classical Lucas parametrization, explains the following normal form and its interpretation by three progressions. We include its short proof to fix the conventions; the integral image-lattice refinement follows separately.

Theorem 1 (Seven-line normal form). *Up to the dihedral symmetry of the grid, every 3×3 integer array in which exactly seven of the eight ordinary lines share a common sum has the antidiagonal as its failed line and is exactly*

$$\begin{pmatrix} A & P + D & E - D \\ P - D & E & A + D \\ E + D & A - D & P \end{pmatrix},$$

three arithmetic progressions of one common difference D ; the seven lines sum to $A + E + P$ and the failed diagonal to $3E$. Consequently an $(8, 7)$ array consists of two complete square progressions and one progression containing exactly two squares. Conversely, the displayed array has exactly seven equal line sums whenever $A + P \neq 2E$.

Proof. The failed line cannot be a row: the three columns already force the total $3S$, so if two rows have sum S the third does as well. The same argument excludes a failed column. After a grid symmetry the failed line is the antidiagonal. Set the principal diagonal entries to A, E, P , so the common sum is $S = A + E + P$, and write the middle-right entry as $A + D$. The middle row and last column give the entries $P - D$ and $E - D$, respectively. The first row, first column and middle column then give $P + D, E + D$ and $A - D$. This is the displayed matrix, with integer D and failed sum $3E$. Its entries split into

$$(A - D, A, A + D), \quad (P - D, P, P + D), \quad (E - D, E, E + D).$$

Exactly eight square entries therefore leave two full square progressions and a third progression containing exactly two squares. $\square$

For distinct-entry arrays $D \neq 0$. We use $d = |D| > 0$ for the positively oriented difference of a square progression. Reversing the sign of D is accounted for by a grid symmetry. A *transversal*

configuration is a triple of positive square progressions of common difference d , with all nine entries distinct. It is *full-magic* when its centres satisfy $s_1^2 + s_2^2 = 2s_0^2$, where index 0 is the central progression. Whenever the finiteness or support laws refer to a configuration, this interaction condition is explicitly imposed. Prime supports concern d or $2d$, never the prime factors of the nine entries.

Remark 1 (Where full magic lives). In the normal form, the array is fully magic if and only if the antidiagonal sum equals the common sum, that is $3E = A + E + P$, equivalently

$$A + P = 2E :$$

the three centres themselves must form a three-term arithmetic progression. For an all-squares array this is a *fourth* square progression, with its own common difference $E - A$ — never equal to $\pm D$, since $E = A + D$ would repeat the centre at positions $(2, 2)$ and $(2, 3)$, while $E = A - D$ would repeat it at $(2, 2)$ and $(3, 2)$, against nine-way distinctness — and this single extra condition is exactly where the elliptic interaction of Section 5 lives. The classical one-parameter families of nine-square arrays with seven equal sums (Lucas [Luc76]; Sallows's record example [Sal97]) sit inside this normal form, as does the general parametrization of 3×3 magic squares going back to Lucas [Luc91].

2.1 The integral image lattice

Proposition 1 (Integral line-sum lattice). *Let $\Lambda : \mathbb{Z}^9 \rightarrow \mathbb{Z}^8$ send $X = (a, b, c; d, e, f; g, h, i)$ to $(R_1, R_2, R_3, C_1, C_2, C_3, D_+, D_-)$, its three row sums, three column sums and two diagonal sums. Then*

$$\text{SNF}(\Lambda) = \text{diag}(1, 1, 1, 1, 1, 1, 3, 0), \quad \text{coker}(\Lambda) \cong \mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}.$$

An integer vector of eight prescribed line sums lies in the image if and only if

$$R_1 + R_2 + R_3 = C_1 + C_2 + C_3, \quad R_1 + R_3 \equiv C_2 + D_+ + D_- \pmod{3}.$$

Consequently Λ has rank 6 over $\mathbb{F}_3$ and rank 7 over $\mathbb{F}_p$ for every prime $p \neq 3$.

Proof. In the entry order $(a, b, c, d, e, f, g, h, i)$, the matrix of Λ is

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

The primitive left relation $-R_1 - R_2 - R_3 + C_1 + C_2 + C_3 = 0$ bounds the rank by 7. The minor on rows $(R_1, R_2, R_3, C_1, C_2, D_+, D_-)$ and columns (a, b, c, d, e, f, g) is -3 , whereas the minor on rows $(R_1, R_2, R_3, C_1, C_2, D_+)$ and columns (a, b, c, d, f, g) is -1 . Moreover

$$R_1 + R_3 - C_2 - D_+ - D_- = -3e,$$

so modulo 3 the rank is at most 6 and every 7-minor is divisible by 3. The sixth and seventh determinantal divisors are therefore 1 and 3, giving the displayed Smith form. The two stated conditions annihilate the image and define a surjective map $\mathbb{Z}^8 \rightarrow \mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ (use the R_2 and D_+ coordinate vectors); its kernel has the same quotient as $\text{im } \Lambda$, so the two coincide. Reduction of the Smith form gives the rank statement. $\square$

2.2 Retraction bookkeeping

Lemma 1 (Retraction bookkeeping). *Let a fully magic 3×3 array have common line sum M . (i) Subtracting the same t from the three cells of one long diagonal leaves the three rows, the three columns and the other diagonal with sum $M - t$, while the selected diagonal has sum $M - 3t$; the exact line defect is $2t$. (ii) More generally, subtracting τ from the three cells of a permutation transversal T leaves every row and every column with sum $M - \tau$; if T meets the two diagonals in j and k cells, their sums become $M - j\tau$ and $M - k\tau$. (iii) Conversely, if exactly one diagonal of a 3×3 array fails while the other seven lines share a common sum, adding half the defect to each cell of the failed diagonal restores a fully magic array.*

Proof. (i) Every row and every column contains exactly one modified cell, and the other diagonal contains exactly one modified cell, while the selected diagonal contains all three; the sums follow, and the defect is $(M - t) - (M - 3t) = 2t$. Nothing here uses squareness or any property of a particular family. (ii) A permutation transversal meets every row and every column in exactly one cell, and meets the diagonals in j resp. k cells by definition of j, k ; the same count gives the sums. (iii) By Theorem 1 the array is the displayed normal form, with seven-line sum $A + E + P$, failed antidiagonal sum $3E$, and defect $\delta = A + P - 2E$. Each row and column contains exactly one antidiagonal cell, and the principal diagonal contains one (the centre), so adding $\delta/2$ to the three antidiagonal cells raises all seven line sums to $A + E + P + \delta/2$ and the antidiagonal to $3E + 3\delta/2$; both equal $\frac{3}{2}(A + P)$, so all eight lines agree. $\square$

Remark 2 (Census normalization). The bounded census of Appendix A quotients by the geometric D_4 action and measures height by the largest square root among the eight square entries; its exact minimum statements are supporting propositions at their frozen bound and are never used as evidence for an unbounded claim.

3 Prime-support exclusions and exact catalogues

We first deduce a support-size exclusion from Aebi's classification, then give an effective classification for three specified prime sets. Both arguments use primitive right triangles, but their completeness inputs are different. Neither depends on the elliptic interaction model or the five-term equation developed later.

3.1 At most three prime factors

Theorem 2 (Complete three-prime exclusion). *Consequence of Aebi's classification. For any positive integer d with at most three distinct prime divisors, there is at most one positive integer triple $r < s < t$ such that $s^2 - r^2 = t^2 - s^2 = d$. No primitivity assumption is required. In particular, neither a $(9, 7)$ array nor a fully magic square of nine distinct positive integer squares can have such a transversal difference. The primitive Pythagorean-triangle classification is Aebi's; the conclusion follows by comparing its area squareclasses.*

Proof of Theorem 2. Divide the roots of a positive square progression by their gcd g . The primitive roots r', s', t' are all odd (reduce modulo 4), and $U = (t' - r')/2$, $V = (t' + r')/2$, s' form a primitive right triangle: $U^2 + V^2 = s'^2$, $\gcd(U, V) = 1$. Its area $A = UV/2$ satisfies $d/4 = Ag^2$, so its prime support has at most three elements.

Aebi's theorem and closing remark [Aeb26] classify all such primitive triangles, up to exchanging the legs. Their areas and squarefree parts are, respectively,

A	6	30	60	180	84	504	1224
$\text{sf}(A)$	6	30	15	5	21	14	34

and each area has exactly one primitive triangle. The squarefree parts are distinct. If two positive progressions had the same d , the ratio of their primitive areas would be a rational square, forcing the same area and the same g . The triangle then recovers the unique roots $g|U - V|, gs', g(U + V)$, so the progressions coincide. This proves the statement and its array consequences. The bound on the number of primes cannot be raised to four for this uniqueness statement: at $d = 840$ the two triples $(1, 29, 41)$ and $(23, 37, 47)$ are distinct examples. $\square$

3.2 The fixed-support classification

For $S_B = \{p \text{ prime} : p \leq B\}$, with $B = 7, 13, 19$, we solve a two-term S -unit problem, reconstruct the primitive triangles and their normalized square progressions, and test every centre interaction. The result is a complete classification for the specified supports, with unrestricted exponents.

Theorem 3 (Smooth differences and a complete $(9, 7)$ atlas). ***Computer-assisted, using imported complete S -unit totals.** Let S_B be the primes at most B . (i) There is no fully magic 3×3 square of nine distinct positive integer squares whose transversal difference d satisfies $\text{supp}(d) \subseteq S_{19}$. (ii) The $(9, 7)$ arrays with $\text{supp}(|D|) \subseteq S_B$, modulo D_4 and common rational-square scaling, have exactly 3, 45, 384 classes for $B = 7, 13, 19$, respectively, represented by primitive positive roots. (iii) For $B = 19$, the sharp lower bound for $\eta = |A + P - 2E|/|D|$ is $347984603/9837828000$, attained by exactly one class, already present for $B = 13$. These are unbounded-exponent statements: completeness uses Theorem A of von Känel–Matschke, not a search cutoff.*

3.3 The imported total and its normalization

Let $\mathcal{A}(S)$ be the set of integer triples

$$0 < a \leq b < c, \quad a + b = c, \quad \gcd(a, b) = 1, \quad \text{supp}(abc) \subseteq S.$$

Theorem A of von Känel and Matschke [vKM16], printed p. 7, counts solutions of $x + y = 1$ in rational S -units modulo the six usual symmetries. Each orbit has exactly one representative $0 < x \leq 1/2$; writing it as $x = a/c$, $1 - x = b/c$ gives exactly one element of $\mathcal{A}(S)$. The exceptional orbit represented by $(1, 1, 2)$ is counted once. Thus their entries for the first four, six and eight primes give

$$|\mathcal{A}(S_7)| = 63, \quad |\mathcal{A}(S_{13})| = 545, \quad |\mathcal{A}(S_{19})| = 3649.$$

The first two totals were already obtained by de Weger, as credited in [vKM16]. We import only these three totals, not a conjecture comparing different supports of the same size.

The supplied producer finds 3649 distinct valid triples with $c \leq 10^{10}$. The separate verifier checks integrality, positivity, ordering, coprimality, addition and prime support for every row, and checks the imported total. A subset of a finite set with the same cardinality is the entire set: this is the completeness argument, with *no remaining height restriction*. The production bound alone would prove no such claim. The witnesses are regenerated locally, not copied from the authors' data files; their source theorem remains a mathematical dependency.

3.4 From additive triples to square progressions

Lemma 2 (Exact triangle reconstruction). *Primitive positive integer right triangles, up to swapping the legs, whose area is supported on S correspond bijectively to pairs $m > k > 0$ with $\gcd(m, k) = 1$, $m + k$ odd, and*

$$(\min(k, m - k), \max(k, m - k), m) \in \mathcal{A}(S), \quad (k, m, m + k) \in \mathcal{A}(S).$$

Their legs, hypotenuse and area are

$$\{a, b\} = \{2mk, m^2 - k^2\}, \quad c = m^2 + k^2, \quad A = mk(m - k)(m + k).$$

Proof. Euclid's parametrization is unique with these coprimality and parity conditions. Its area is supported on S exactly when all four positive factors $m, k, m - k, m + k$ are supported on S . The two displayed additive triples express precisely these conditions; their coprimality follows from $\gcd(m, k) = 1$. Conversely, membership and odd $m + k$ give Euclid's conditions and the required area. This proves both directions, not just a procedure for generating examples. $\square$

Write $A = nh^2$ with n squarefree. The corresponding normalized square progression has positive rational roots

$$\left(\frac{|a - b|}{h}, \frac{c}{h}, \frac{a + b}{h} \right), \quad d = 4n, \quad \sigma = \frac{c^2}{h^2} > 4n.$$

Conversely, divide any positive integer-root square progression by the gcd g of its roots. Write the divided roots as r', s', t' ; they are odd, and $((t' - r')/2, (t' + r')/2, s')$ is a primitive right triangle of area $d/(4g^2)$. Thus $d/4 = n(hg)^2$. Progressions of the same difference necessarily have the same n and common scale $\lambda = hg$, so their actual centres are $\lambda^2\sigma$. Consequently the full-magic interaction is exactly

$$\sigma_1 + \sigma_2 = 2\sigma_0.$$

All denominators of the normalized roots divide h and are supported on S . Clearing them and dividing by the common root gcd therefore gives every root-primitive integral array without introducing a prime outside S into its difference.

3.5 The complete finite tests

Proof of Theorem 3. The complete additive lists and Lemma 2 give the following exact counts. A midpoint test checks one unordered pair of distinct centres within the same squarefree area class and asks whether their average is another centre in that class.

B	$ \mathcal{A}(S_B) $	triangles	classes	pairs	hits	$(9, 7)/D_4$ classes
7	63	12	9	4	0	3
13	545	62	39	36	0	45
19	3649	232	133	180	0	384

Every number in this table is recomputed with integer and rational arithmetic by the companion's exact checker:

`scripts/verify_smooth_support.py.`

In particular, the normalized centres within each class are distinct and no midpoint test succeeds. By the preceding equivalence, no fully magic square in the stated domain exists. This establishes part (i).

For part (ii), group the triangles by n and select three distinct triangles in one group. The three normalized progressions give nine candidate entries. Reject nonpositive or repeated entries; for these catalogues none of the 1, 15, 128 unordered triples is rejected. Choose one of the three centres as the central cell E , use the other two as A, P in the normal form, and set $D = 4n$. The absence of midpoint hits ensures exactly seven equal line sums. Clear root denominators, divide by the root gcd, and take the lexicographically least positive-root array in its eight-element D_4 orbit.

These operations are exhaustive by the normal form, which also holds when all eight lines agree. For an array with exactly one failed line, orienting that diagonal as the antidiagonal leaves precisely the choices of interchanging A, P and changing the sign of D . Those choices are already identified by D_4 ; the central cell is fixed by every grid symmetry. Therefore the three choices of E are inequivalent. The unique primitive normalization of a positive rational-root array also removes all common rational-square scaling. The verifier reconstructs and compares all 3, 45, 384 representatives, establishing the classification.

Finally, for each representative compute the seven-line sum M , the failed sum F , and

$$\eta = \frac{|M - F|}{|D|} = \frac{|A + P - 2E|}{|D|}.$$

This quotient is invariant under D_4 and common square scaling. Exact comparison over the complete lists gives minima 1727/1120 for S_7 and 347984603/9837828000 for both S_{13} and S_{19} , each attained by one class. For the latter, a primitive root representative (the *entries of the array are its squares*) is

$$\begin{pmatrix} 87942 & 140595 & 157690 \\ 186290 & 132558 & 9555 \\ 99645 & 122590 & 165558 \end{pmatrix}.$$

It has $|D| = 9837828000$, $M = 52366885489$, $F = 52714870092$, and $M - F = -347984603$. These identities and uniqueness are checked from the full atlas, proving part (iii). No floating-point comparison is used. $\square$

For S_7 the unique unordered triple, after common root-primitive normalization, has roots (2, 58, 82), (46, 74, 94) and (97, 113, 127) and common difference 3360. Houston [Hou16b] already displays exactly these three progressions: his elliptic parameter has difference 210, and the displayed integral roots give difference 3360. The contribution here is their place in the complete fixed-support classification. Their three choices of the central progression account for the three classes. The atlas is for (9, 7) arrays; it is separate from the bounded (8, 7) census of Appendix A. Also, η is normalized by the progression difference, not by the magic sum: it is not a percentage error or a comparison with unrestricted near-miss records.

3.6 Consequences and limits

Every hypothetical fully magic square of distinct positive integer squares must have a prime at least 23 in each of its two transversal differences. Indeed, write its normal form as $M(E - t, E, E + t; d)$, where M denotes the matrix in Theorem 1. A clockwise quarter-turn gives $M(E + d, E, E - d; t)$. Both differences are nonzero, and applying part (i) in each orientation proves the assertion. The two large primes need not be different. These are primes of the *differences*, not of the entries.

In particular, the fifteen supports consisting of 2, 3 and two distinct primes chosen from $S_{19} \setminus \{2, 3\}$ are excluded. This does not close arbitrary four-prime supports, bound the number of support primes, or settle the general magic-square-of-squares problem. The general fixed-squareclass finiteness result of Section 6 remains non-effective; the present enumeration is obtained through the separate fixed-support route and has been executed only for the declared catalogues.

4 Constructions of (8, 7) near misses

We use three classical starting points in different ways. Retraction of the Bremner–Wesolowski carriers and an elliptic-orbit lift each yield an infinite family. The intervening difference-840 bridge admits a complete finite classification. The statement below records these distinct results; the proof of each part follows in its own subsection.

Theorem 4 (Two infinite constructions and a finite bridge). (i) *Subtracting $6z_n^2$ from the antidiagonal of the n -th catalogued Bremner–Wesolowski carrier yields, for every $n \geq 1$, an (8, 7) array with explicit square entries and defect $12z_n^2$; every odd-indexed member is root-primitive.* (ii) *For fixed n , all positive shifts $0 < t < C$ closing at least one antidiagonal pair to squares are classified by same-parity factor pairs of Δ and 2Δ , and the least such shift is 9 for $n = 1$ and $6z_n^2$ for every $n \geq 2$; larger pair-closing shifts can also occur.* (iii) *In the type-6.VI frame the fixed bridge $(23^2, 37^2, 47^2)$ of difference 840 admits exactly the tautological completion $(23, 37, 47)$ and one further completion $(79, 65, 89)$; coupled with Robertson’s progression $(1^2, 29^2, 41^2)$ it produces an explicit (8, 7) array.* (iv) *The non-torsion orbit of $P = (841, -1189)$ on $E_{840}: Y^2 = X^3 - 840^2 X$ re-enters the admissible window infinitely often and yields infinitely many pairwise square-scale/ D_4 -inequivalent, entry-primitive (8, 7) arrays from Bremner’s Configuration VI. More precisely, the admissible indices have natural density*

$$\delta = 1 - I_{(840/4225)^2} \left(\frac{1}{4}, \frac{1}{2} \right) = 0.6585271498271213 \dots$$

4.1 The retraction family and its least-shift law

The carrier is the six-square fully magic family recorded on Boyer’s catalogue pages [Boy] and attributed there to Wesolowski; the catalogue states explicitly that the recurrence below “was not given by Arkadiusz”, and the $n = 1$ grid already occurs as Configuration II of Bremner [Bre01], so we speak of the Bremner/Wesolowski carrier. With $(x_0, y_0, z_0) = (3, 1, 1)$ and

$$x_{n+1} = 3x_n + 2z_n, \quad y_{n+1} = x_n, \quad z_{n+1} = x_n + z_n,$$

the carrier W_n is fully magic with centre $E = 4z^4 + 5z^2 + 1$ and common sum $3E$, where $(x, y, z) = (x_n, y_n, z_n)$. Subtracting $t = 6z^2$ from its antidiagonal and using the recurrence identities

$$x^2 + y^2 z^2 - 6z^2 = (yz + 1)^2, \quad x^2 z^2 + y^2 - 6z^2 = (xz - 1)^2, \quad \frac{(x^2 + y^2)(z^2 + 1)}{2} - 6z^2 = 4z^4 - z^2 + 1$$

produces

$$N_n = \begin{pmatrix} (xy - z)^2 & (xz + y)^2 & (yz + 1)^2 \\ (x + yz)^2 & 4z^4 - z^2 + 1 & (xz - y)^2 \\ (xz - 1)^2 & (yz - x)^2 & (xy + z)^2 \end{pmatrix}.$$

Proof of Theorem 4(i). By Lemma 1(i) the seven unselected lines of N_n share the sum $3E - 6z^2 = 3(4z^4 + 3z^2 + 1)$, the antidiagonal sums to $3E - 18z^2 = 3(4z^4 - z^2 + 1)$, and the defect is $12z^2$. For $z > 1$,

$$(2z^2 - 1)^2 < 4z^4 - z^2 + 1 < (2z^2)^2,$$

so the centre is not a square. The eight positive roots are, in increasing order,

$$yz - x < yz + 1 < x + yz < xy - z < xy + z < xz - y < xz - 1 < xz + y :$$

positivity of the first follows from $yz - x = y(z - 1) - 2z \geq 3(z - 1) - 2z = z - 3 > 0$, most gaps are immediate, and the two non-obvious ones reduce, through the recurrence invariants, to

$$(xy - z) - (x + yz) = 2z^2 - yz - y - 3z + 1 > 0, \quad (xz - y) - (xy + z) = (y - 1)(z - 1) - 2 > 0,$$

the first bounded below by $(z - 1)(z - 2) > 0$ using $y \leq z - 1$, the second at least 4. Hence all nine entries are positive and distinct. For root-primitivity: an integer dividing all eight roots divides both $(yz + 1) - (yz - x) = x + 1$ and $(x + yz) - (yz + 1) = x - 1$, hence divides 2; the recurrence keeps x, y odd and toggles the parity of z , so for odd n all eight roots are odd and their gcd is 1. There are infinitely many odd indices. $\square$

Proposition 2 (Complete pair-closing classification). *Fix n and write the retracted antidiagonal entries of the carrier as $C < E < G$ with common gap Δ . Every positive shift $0 < t < C$ for which at least two of $C - t$, $E - t$, $G - t$ are squares arises from a same-parity factor pair, and every retained pair works: for branch CE , from $de = \Delta$, $0 < d < e$, $d \equiv e \pmod{2}$, via $u = (e - d)/2$, $w = (e + d)/2$, $t = C - u^2$; for branch EG , from $de = \Delta$ via $w = (e - d)/2$, $v = (e + d)/2$, $t = E - w^2$; for branch CG from $de = 2\Delta$ via $u = (e - d)/2$, $v = (e + d)/2$, $t = C - u^2$.*

Proof. For CE , subtracting the two square equations gives $w^2 - u^2 = \Delta = (w - u)(w + u)$, so $d = w - u$ and $e = w + u$ have the stated product and parity, and the inverse formulas recover u, w . The other branches are identical with difference Δ or 2Δ . This is a divisor classification, not a scan over t . $\square$

Proposition 3 (Least shift). *For every $n \geq 2$, $t_0 = 6z^2$ is the least positive shift closing at least one pair; at $n = 1$ the least is 9. Hence $t_{\min}(1) = 9$ and $t_{\min}(n) = 6z_n^2$ for $n \geq 2$. Leastness does not mean uniqueness: larger nondegenerate closing shifts exist and are recorded exactly.*

Proof frame. For $n \geq 2$ the recurrence on (y, z) is $(y, z) \mapsto (y + 2z, y + 3z)$, starting at $(11, 15)$, and direct induction gives

$$z > y, \quad 2y > z, \quad 5y > 3z, \quad 5z > 6y, \quad 8y > 5z, \quad z \geq 15.$$

Assume $0 < t < t_0$ closes a pair and put $h = t_0 - t$, $0 < h < 6z^2$. At t_0 the three entries are $C - t_0 = Y^2$ with $Y = yz + 1$, $E - t_0 = 4z^4 - z^2 + 1$, and $G - t_0 = X^2$ with $X = xz - 1$; each entry can be square inside the short interval only along an explicit one-parameter list (for the C entry, $C - t = (Y + a)^2$ forces $h = a(2Y + a)$, and similarly for E and G), and the listed windows are excluded pairwise by the induction inequalities. The complete three-case interval analysis, with its exact certificate, is reproduced in Appendix B. At $n = 1$, $(C, E, G) = (265, 1105, 1945)$: for $1 \leq t \leq 8$ each retracted entry lies strictly between consecutive squares ($16^2 < C - t < 17^2$, $33^2 < E - t < 34^2$, $44^2 < G - t < 45^2$), so no pair closes below $t = 9$, while $t = 9$ gives $(256, 1096, 1936) = (16^2, 1096, 44^2)$. $\square$

Parts (i)–(ii) of Theorem 4 are proved.

4.2 The bridge completion in type 6.VI

The catalogued carrier type 6:6 (6.VI here) is normalized, after the dihedral action, as

$$M(A, N, Z) = \begin{pmatrix} 2N - Z & A & N - A + Z \\ 2Z - A & N & 2N - 2Z + A \\ N + A - Z & 2N - A & Z \end{pmatrix},$$

with square cells $A = a^2$, $N = n^2$, $Z = z^2$, $B = 2N - A = b^2$, $D = 2Z - A = d^2$, $C = 2N - D = c^2$. The type taxonomy and database records are Fituvalu's [Fit]; the fixed numerical carrier used below is historically Bremner's Configuration VI [Bre01].

Proposition 4 (The four-progression network). *The six source squares satisfy $a^2 + b^2 = 2n^2$, $d^2 + c^2 = 2n^2$ and $d^2 + a^2 = 2z^2$, so the source contains the three square progressions (a^2, n^2, b^2) , (d^2, n^2, c^2) , (d^2, z^2, a^2) . With $\delta = a^2 - z^2 = z^2 - d^2 > 0$, the selected antidiagonal is identically $(n^2 - \delta, n^2, n^2 + \delta)$, and subtracting $t > 0$ closes it to squares (x^2, y^2, w^2) exactly when*

$$x^2 = y^2 - \delta, \quad w^2 = y^2 + \delta, \quad t = n^2 - y^2 > 0 :$$

a fourth square progression, with the difference of the bridge (d^2, z^2, a^2) .

Proof. Substituting $D = 2Z - A$ into the normalized array gives $A - Z = Z - D = \delta$, and the antidiagonal entries become $N - \delta$, N , $N + \delta$; a common subtraction preserves their differences, and the closure conditions are the displayed ones. $\square$

Proposition 5 (Complete bridge-completion classification). *Fix an increasing bridge progression (d^2, z^2, a^2) of difference δ . Every positive type-6.VI completion arises from a same-parity factor pair*

$$uv = 2\delta, \quad 0 < u < v, \quad u \equiv v \pmod{2},$$

via $b = (v - u)/2$, $c = (v + u)/2$, $n^2 = (a^2 + b^2)/2$, a pair being retained exactly when n^2 is an integer square; then $d^2 + c^2 = a^2 + b^2 = 2n^2$ automatically. The pair $(u, v) = (a - d, a + d)$ always returns the tautological completion $(b, n, c) = (d, z, a)$, which repeats square cells and is not admissible. Positivity, nine-way distinctness and the six-square mask are separate final checks.

Proof. Subtracting the two centre relations gives $c^2 - b^2 = a^2 - d^2 = 2\delta$, so $u = c - b$, $v = c + b$ have the stated product and parity, and inversion recovers b, c ; conversely any retained pair satisfies $c^2 - b^2 = 2\delta = a^2 - d^2$, so the two expressions for $2n^2$ agree. $\square$

Proof of Theorem 4(iii). For the fixed bridge $(23^2, 37^2, 47^2)$, $\delta = 840$: running Proposition 5 over the finitely many same-parity factor pairs of $2\delta = 1680$ retains exactly the tautological $(23, 37, 47)$ and the unique admissible completion $(79, 65, 89)$ (exact finite check; certificate in Appendix B). Coupling the resulting carrier with Robertson's classical progression $(1^2, 29^2, 41^2)$ of the same difference [Rob96] retracts the antidiagonal by $\tau = 65^2 - 29^2 = 3384$ and yields the explicit $(8, 7)$ array

$$\begin{pmatrix} 7081 & 2209 & 1 \\ 529 & 841 & 7921 \\ 1681 & 6241 & 1369 \end{pmatrix}$$

with seven-line sum 9291, failed antidiagonal sum 2523, and unique nonsquare 7081. $\square$

4.3 The elliptic-orbit lift

The fixed carrier is

$$M_0 = \begin{pmatrix} 7081 & 2209 & 3385 \\ 529 & 4225 & 7921 \\ 5065 & 6241 & 1369 \end{pmatrix},$$

Bremner's Configuration VI [Bre01], in the normalized orientation of Fituvalu database row 340545 [Fit], is fully magic with sum 12675. Its six square entries have outside roots

$$\{23, 37, 47, 79, 89\},$$

and its selected antidiagonal is $(65^2 - 840, 65^2, 65^2 + 840)$. Let

$$E: Y^2 = X^3 - 840^2 X, \quad Q = (1960, 78400), \quad P = 2Q = (841, -1189).$$

Lemma 3 (Doubling produces a square progression). *For $R = (u, v) \in E(\mathbb{Q})$ with $v \neq 0$,*

$$x(2R) = \left(\frac{u^2 + 840^2}{2v} \right)^2, \quad x(2R) \mp 840 = \left(\frac{u^2 \mp 1680u - 840^2}{2v} \right)^2,$$

so every finite point of $2E(\mathbb{Q})$ supplies three rational squares in arithmetic progression with common difference 840 — the classical congruent-number correspondence (cf. [Rob96]), in the exact convention used by the lift.

Lemma 4 (Fixed-carrier lift). *Let $R \in 2E(\mathbb{Q})$ with $840 < x(R) < 65^2$. Write the positive roots of $x(R) - 840$, $x(R)$, $x(R) + 840$ in lowest terms and let λ be the least common multiple of their denominators. Scale M_0 by λ^2 and subtract $\tau = \lambda^2(65^2 - x(R))$ from the selected antidiagonal; the antidiagonal becomes the three cleared squares. If the three new roots avoid $\{23, 37, 47, 79, 89\}$, the output is positive with nine distinct entries and exactly eight squares, the unique nonsquare being $7081\lambda^2$ — and it is entry-primitive: a common prime p of all entries would divide λ ; choosing a reduced root denominator of maximal p -adic valuation, its cleared numerator stays prime to p , and its square blocks p . The argument uses minimality of λ and fails for an arbitrary common multiple.*

Lemma 5 (Q is non-torsion). *$P = 2Q = (841, -1189)$ is integral with $1189 = 29 \cdot 41$, while $4 \cdot 840^6$ has only the prime divisors $2, 3, 5, 7$; by the Lutz–Nagell criterion [Sil09, Chap. VII, Prop. 3.1] an integral torsion point with nonzero ordinate would satisfy $Y^2 \mid 4 \cdot 840^6$, so P is non-torsion, hence so is Q . Independently, $\#E(\mathbb{F}_{11}) = 12$ and $\#E(\mathbb{F}_{13}) = 20$ at good primes bound the rational torsion order by $\gcd(12, 20) = 4$, and P is neither infinity nor rational two-torsion.*

Proof of Theorem 4(iv). P lies in the identity component $E(\mathbb{R})^0$, a circle group, and is non-torsion, so its positive multiples are dense there. The open set $U = \{R \in E(\mathbb{R})^0 : 840 < x(R) < 4225\}$ contains P ; removing the finite set where one of the three roots equals one of $23, 37, 47, 79, 89$ leaves a nonempty open set V — the roots at P itself are $(1, 29, 41)$. Density gives infinitely many positive multiples $mP = 2mQ$ in V , and Lemma 4, always with the least common multiple, turns each return into a primitive positive $(8, 7)$ array. If the outputs at indices m, n were related by a D_4 action and a positive square scale ρ^2 , the unique nonsquare entries would force $7081\lambda_n^2 = \rho^2 7081\lambda_m^2$, so $\rho = \lambda_n/\lambda_m$; a D_4 action fixes the centre, so comparing centres gives $x(mP) = x(nP)$, hence $mP = \pm nP$, and non-torsion with $m, n > 0$ forces $m = n$. Distinct return indices therefore give distinct square-scale/ D_4 classes. Two exact witnesses, including the $m = 3$ return, are frozen in the companion repository. $\square$

Corollary 1 (Positive-density orbit lift). *Let $A(N)$ count the indices $1 \leq m \leq N$ for which mP lies in the admissible set V of the preceding proof. Then*

$$A(N) = \delta N + o(N), \quad \delta = \frac{\int_{840}^{4225} dx / \sqrt{x^3 - 840^2 x}}{\int_{840}^{\infty} dx / \sqrt{x^3 - 840^2 x}} = 1 - I_{(840/4225)^2} \left(\frac{1}{4}, \frac{1}{2} \right),$$

and

$$\delta = 0.6585271498271213 \dots$$

The same asymptotic counts the square-scale/ D_4 -inequivalent primitive $(8, 7)$ classes produced among $P, 2P, \dots, NP$.

Proof. The multiples of the non-torsion point P are equidistributed in the circle group $E(\mathbb{R})^0$ with respect to normalized Haar measure; see Cowan [Cow20] for the natural-density formulation for real elliptic curves. The boundary of $840 < x < 4225$ has measure zero, and deleting the finitely many collision points changes neither measure nor density. On the identity component the invariant measure is proportional to $dx / \sqrt{x^3 - 840^2 x}$, giving the ratio above. Substitution $x = 840/t^2$ changes it to

$$\frac{\int_{\sqrt{840/4225}}^1 dt / \sqrt{1 - t^4}}{\int_0^1 dt / \sqrt{1 - t^4}},$$

which is the stated regularized incomplete-beta value. The final counting claim follows from the inequivalence argument in the preceding proof. $\square$

Remark 3 (The universal lemniscatic law behind δ). The special-function value in Corollary 1 is not peculiar to 840. For $d > 0$ let

$$E_d : \quad Y^2 = X^3 - d^2 X.$$

If $R = (X, Y)$ is distributed according to normalized Haar probability on $E_d(\mathbb{R})^0$ and $U = (d/X)^2$, then the push-forward density is

$$d\mu_U(u) = \frac{u^{-3/4}(1-u)^{-1/2}}{B(\frac{1}{4}, \frac{1}{2})} du, \quad 0 < u < 1;$$

equivalently, $U \sim \text{Beta}(\frac{1}{4}, \frac{1}{2})$. Indeed, $X = du^{-1/2}$ sends the invariant differential $dX / \sqrt{X^3 - d^2 X}$ to a constant multiple of $u^{-3/4}(1-u)^{-1/2} du$. Consequently, for every $B > d$,

$$\mu\{R \in E_d(\mathbb{R})^0 : d < X(R) < B\} = 1 - I_{(d/B)^2} \left(\frac{1}{4}, \frac{1}{2} \right).$$

The normalizing integral is the classical lemniscatic value

$$\int_0^1 \frac{dt}{\sqrt{1 - t^4}} = \frac{\Gamma(1/4)^2}{4\sqrt{2}\pi}$$

recorded in the NIST Digital Library of Mathematical Functions [DLMF, §19.20(i), Eq. (19.20.2)]. This universality reflects $j(E_d) = 1728$: after real scaling, every curve in the family has the same

normalized invariant measure, so the probability depends only on B/d . For a non-torsion point in $E_d(\mathbb{R})^0$, equidistribution transfers this probability to orbit indices. A point on the other real component requires separate bookkeeping for its even and odd multiples. Thus the remark explains the density constant but adds no new arithmetic input to the corollary.

Remark 4 (From constructions to the wall). Part (iv) reaches the $(8, 7)$ metric infinitely often from one carrier. Closing the eighth line of the same transversal frame is the interaction problem of Section 5: by Remark 1 it demands a fourth square progression through the centres, and that single extra condition changes the landscape from constructive to arithmetic.

5 The interaction surface

Houston's formulation [Hou16b] identifies full magic with three same-difference square progressions whose centres themselves form an arithmetic progression. We first retain the fixed difference 840 of the preceding construction, then allow the difference to vary and remove the common scale of the roots. These are different arithmetic models. The general setting of linear coordinate relations in E^3 is studied by Caro and García-Fritz [CGF25]; their finiteness theorem is specialized in Section 6.

Theorem 5 (The interaction surface). (i) *After independent common subtractions τ_0, τ_1, τ_2 on the three difference-840 transversals of the fixed carrier, all rows and columns remain equal and the two diagonals agree if and only if $\tau_1 + \tau_2 = 2\tau_0$. Requiring square entries gives $x_i \in x(2E_{840}(\mathbb{Q}))$ and the interaction $x_1 + x_2 = 2x_0$.* (ii) *With nonzero common difference d allowed to vary, the signed-root interaction model, modulo common scaling of the roots, has the irreducible surface Y° defined below as its nonzero, distinct-entry locus. Its map to the rational cone $2A^2 = B^2 + C^2$ has generic degree five in squared coordinates and degree forty on the full signed cover; the quintic reconstruction over the rational function field of the cone has Galois group S_5 . For a prescribed difference d_0 , recovering the roots additionally requires a nonzero scale L satisfying*

$$L^2 = \frac{2d_0W}{RS}.$$

The irreducibility assertion concerns Y° , and the Galois-group assertion concerns its quintic reconstruction, without an assertion about this fixed-difference scale locus or the group of the full signed cover.

5.1 The transversal equation

For arbitrary x_0, x_1, x_2 consider

$$\begin{pmatrix} x_1 & x_2 + 840 & x_0 - 840 \\ x_2 - 840 & x_0 & x_1 + 840 \\ x_0 + 840 & x_1 - 840 & x_2 \end{pmatrix}.$$

Every row, every column and the principal diagonal sums to $x_0 + x_1 + x_2$, while the antidiagonal sums to $3x_0$; all eight sums agree if and only if

$$x_1 + x_2 = 2x_0.$$

This proves Theorem 5(i): imposing that each x_i be the x -coordinate of a point of $2E_{840}(\mathbb{Q})$ — the arithmetic condition of Lemma 3 — makes the fully magic completion of the transversal frame exactly one linear equation on E_{840}^3 , hence a surface. The diagonal (P, P, P) is not isolated, and

the membership condition $P_i \in 2E_{840}(\mathbb{Q})$ is arithmetic rather than a further geometric equation over an algebraic closure.

For later use we record the Mordell–Weil data. The minimal model of E_{840} is recorded as LMFDB 705600.vn3 [LMFDB], $Y^2 = X^3 - 44100X$, with rank 2, full rational 2-torsion and free generators $(-84, 1764)$, $(294, 3528)$; transporting by $(x, y) = (4X, 8Y)$ and adjusting by the two-torsion point $T = (-840, 0)$ gives the free basis

$$Q = (1960, 78400), \quad R = (1176, 28224)$$

used throughout, with

$$2Q = (841, -1189), \quad 2R = (1369, -39997).$$

Since doubling kills the rational two-torsion, $2Q$ and $2R$ generate the free part of $2E_{840}(\mathbb{Q})$.

5.2 Normalizing the model with variable difference

Let (r_i^2, s_i^2, t_i^2) , $i = 0, 1, 2$, be three progressions of rational squares with a common difference $d \neq 0$, and retain a choice of their root signs. Put

$$u_i = t_i - r_i, \quad v_i = t_i + r_i.$$

The progression equations and the full-magic interaction become

$$u_i v_i = 2d, \quad u_i^2 + v_i^2 = 4s_i^2, \quad s_1^2 + s_2^2 = 2s_0^2.$$

Conversely these equations recover the roots $r_i = (v_i - u_i)/2$, $t_i = (v_i + u_i)/2$. Since $d \neq 0$, every u_i, v_i is nonzero. Define the scalar coordinates

$$L = u_0, \quad R = \frac{u_1}{u_0}, \quad S = \frac{u_2}{u_0}, \quad W = \frac{u_1 u_2}{2d},$$

$$(A, B, C) = \frac{u_1 u_2}{d u_0} (s_0, s_1, s_2).$$

Here the scalar R is unrelated to the elliptic point denoted by R above. Direct substitution gives the affine variety Y :

$$F = 2R^2 S^2 - (1 + W^2)(R^2 + S^2) + 2W^2 = 0,$$

$$A^2 = W^2 + R^2 S^2, \quad B^2 = W^2 R^2 + S^2, \quad C^2 = W^2 S^2 + R^2.$$

The inverse map, valid for $RSWL \neq 0$, is

$$d = \frac{RSL^2}{2W}, \quad (u_0, u_1, u_2) = (L, RL, SL),$$

$$(v_0, v_1, v_2) = \frac{L}{W}(RS, S, R), \quad (s_0, s_1, s_2) = \frac{L}{2W}(A, B, C).$$

In particular, the three triples of roots are

$$(r_0, s_0, t_0) = \frac{L}{2W}(RS - W, A, RS + W),$$

$$(r_1, s_1, t_1) = \frac{L}{2W}(S - RW, B, S + RW),$$

$$(r_2, s_2, t_2) = \frac{L}{2W}(R - SW, C, R + SW). \tag{1}$$

The radicand equations give the two differences in each progression as $d = RSL^2/(2W)$, and $F = 0$ gives $2A^2 = B^2 + C^2$, hence the interaction. Substituting in either direction recovers every coordinate. Thus the nonzero signed-root model with variable d is isomorphic to the corresponding locus of $Y \times \mathbb{G}_m$, with L the $\mathbb{G}_m$ coordinate. Multiplying all roots by $q \neq 0$ multiplies L by q and d by q^2 , leaving R, S, W, A, B, C unchanged.

For a fixed difference d_0 , this inverse map instead requires

$$L^2 = \frac{2d_0W}{RS}, \quad (x_0, x_1, x_2) = \frac{d_0}{2WRS}(A^2, B^2, C^2). \quad (2)$$

When L is rational, (1) supplies rational roots of $x_i - d_0, x_i, x_i + d_0$, the square-progression data associated with $x(2E_{d_0}(\mathbb{Q}))$. Conversely, these data give the coordinates and scale above. For $d_0 = 840$ the additional equation is $L^2 = 1680W/(RS)$. In particular, an arbitrary rational point of Y does not supply rational roots at difference 840: the quantity on the right must be a rational square. The normalization retains the squareclass $[d] = [RS/(2W)]$ and does not identify all differences with one fixed value.

5.3 The nonzero distinct-entry locus and saturation

The excluded divisors can be read directly from the progression data. Set

$$(\eta_0, \eta_1, \eta_2) = \left(\frac{2W}{RS}, \frac{2WR}{S}, \frac{2WS}{R} \right), \quad f(\eta) = \frac{\eta^2 + 4}{4\eta}.$$

These satisfy $\eta_i = u_i^2/d$ and $s_i^2/d = f(\eta_i)$, so the nine entries divided by d are $f(\eta_i) + \varepsilon$ for $i = 0, 1, 2$ and $\varepsilon \in \{-1, 0, 1\}$. Define Y° by removing $RSWABC = 0$, the numerators of the nine equations $f(\eta_i) + \varepsilon = 0$, and those of the 27 equations

$$f(\eta_i) + \varepsilon = f(\eta_j) + \varepsilon', \quad i < j, \quad \varepsilon, \varepsilon' \in \{-1, 0, 1\}.$$

Within a progression the entries are already distinct because $d \neq 0$. The 27 collision numerators have 30 distinct normalized factors. The coordinate maps preserve these exclusions. Positivity remains a real semialgebraic condition: for positive increasing roots one takes $L, R, S, W, A, B, C > 0$ and $RS > W, S > RW, R > SW$. It is not an additional polynomial saturation.

Proof of Theorem 5(ii), saturation part. As a polynomial in S ,

$$F = (2R^2 - W^2 - 1)S^2 + 2W^2 - (1 + W^2)R^2,$$

whose leading coefficient and constant term are coprime; the coefficient $2R^2 - W^2 - 1$ is squarefree and occurs to odd order in the ratio defining S^2 , so that ratio is not a square in $\mathbb{Q}(R, W)$ and F is irreducible. Over the function field of $F = 0$ put $p_A = W^2 + R^2S^2$, $p_B = W^2R^2 + S^2$, $p_C = W^2S^2 + R^2$. In squared coordinates $r = R^2$, $s = S^2$, $w = W^2$, setting $p_A = 0$ gives $F(r, s, -rs) = (r + s)(rs - 1)$, and on the reduced component $RS = 1$, $W^2 = -1$ the function p_A has generic valuation one while p_B and p_C are generically nonzero; the case $p_B = 0$ reduces to $-2r^2w + rw^2 - r + 2w = 0$, linear in p_B at the generic point of each reduced component with p_A and $p_C = 2p_A$ generically nonzero; the p_C case is symmetric. Hence every nonempty product of the three radicands has odd valuation somewhere and is not a square, so adjoining A, B, C gives a degree-eight extension with integral coordinate ring: the four equations define a single prime component of dimension two. None of the declared zero/collision polynomials vanishes on it, so saturating the prime ideal by their product changes nothing, and Y° is one irreducible surface. $\square$

5.4 The square-progression cone and quintic reconstruction

The four equations imply the exact identity

$$2A^2 = B^2 + C^2 :$$

the three auxiliary roots themselves form a square progression, and the normalized interaction maps Y dominantly to this classical cone. In squared coordinates $a = rs + w$, $b = rw + s$, $c = sw + r$ (so $2a = b + c$), reconstruction away from $r^2 = 1$ is

$$s = \frac{b - ar}{1 - r^2}, \quad w = \frac{a - br}{1 - r^2},$$

where r satisfies the quintic

$$Q(r) = c(1 - r^2)^2 - (b - ar)(a - br) - r(1 - r^2)^2 = 0.$$

Proof of Theorem 5(ii), cone part. The quintic has leading coefficient -1 and the displayed inverse reconstructs s, w from each generic root, so the squared-coordinate map to the cone is generically finite of degree five; each squared triple has eight sign choices of (R, S, W) , giving degree $5 \cdot 8 = 40$ on the full signed cover. For the Galois group over the rational function field of the cone, specialize at the rational cone point $(A, B, C) = (5, 1, 7)$: the reconstruction polynomial

$$-r^5 + 49r^4 + 2r^3 - 123r^2 + 625r + 24$$

is irreducible modulo 7, factors modulo 29 with degree pattern $[2, 1, 1, 1]$, and has nonzero discriminant -24364389070416096000 , so the specialization is separable. Irreducibility supplies a transitive degree-five group containing a five-cycle; the mod-29 pattern supplies a transposition; among transitive subgroups of S_5 this forces the full S_5 , and the generic Galois group contains the specialized one. Since the generic group is itself a subgroup of S_5 , it is exactly S_5 . $\square$

Remark 5 (Scope of the quintic obstruction). The generic S_5 Galois group proves that the cone map has no generic rational section and that its quintic cannot be resolved by a tower of quadratic radicals. It does *not* prove that Y admits no other useful rational map, no height-decreasing correspondence, and no further structure; those remain open research directions, not consequences. These conclusions concern the normalized model with variable difference. No irreducibility or Galois-group conclusion for the fixed- d_0 locus defined by (2) is inferred here.

6 Finiteness, the closed fibre, and the finite-class stop

Theorem 6 (Finiteness and one closed fibre). (i) **Derived from an imported theorem** (Caro–García-Fritz, *Thms. 1.2–1.3*): for fixed $d > 0$, the points of $(2E_d(\mathbb{Q}))^3$ satisfying the interaction, after removing points at infinity, nonpositive and repeated x -coordinates, form a finite set, with the imported bound of shape C^{3r+1} , $r = \text{rank } E_d(\mathbb{Q})$; hence for every fixed squareclass of d only finitely many positive distinct full-magic transversal configurations exist modulo rational-square scaling. The bound is non-effective. (ii) **Computer-assisted** (the decisive rank and torsion enter as a hash-verified frozen transcript of the official Magma Calculator): the fibre $x_0 = 841$ is closed completely — its necessary genus-two curve has a complementary bielliptic quotient of rank zero and full two-torsion, whose four rational points pull back to $t \in \{0, \pm 1, \pm 841, \pm 1681\}$, and positivity leaves only $t = 0$, so the fibre contains no positive nonconstant three-transversal completion.

6.1 Fixed difference and fixed squareclass

Proof of Theorem 6(i). **Imported layer.** Caro and García-Fritz [CGF25] classify the images of nonconstant morphisms $E \rightarrow X_{(c_1, c_2, c_3)} \subset E^3$ cut out by $c_1x(P_1) + c_2x(P_2) + c_3x(P_3) = 0$ (their Theorem 1.2, four types) and bound, via Uniform Mordell–Lang, the intersection of the complement of those curves with any finite-rank subgroup of E^3 (their Theorem 1.3). Both are imported; nothing below re-proves them.

Exact specialization. Over $\mathbb{C}$ every E_d has $j = 1728$, $\text{End}(E_d) = \mathbb{Z}[i]$, unit squares ± 1 ; take $(c_0, c_1, c_2) = (-2, 1, 1)$. Type (a) has two coordinates at infinity, so no finite progression centre. In type (b) the cancellation $c_i + u^2c_j = 0$ forces the moving pair to be the equal-coefficient pair $(1, 1)$ with $u^2 = -1$, and the constant coordinate (the -2 slot) satisfies, in the Weierstrass convention of [CGF25] with quadratic coefficient $A = 0$, $x(R) = -A(c_i + c_j)/(3c_k) = 0$: the two-torsion point $(0, 0)$, whose square progression data is nonpositive or repeated, disjoint from the positive distinct locus. In type (c) the unit condition $-2 + 1/u^2 + 1/v^2 = 0$ with inverse unit squares ± 1 forces $u^2 = v^2 = 1$, i.e. three equal x -coordinates — the repeated diagonal. Type (d) is absent twice over: its exceptional curve has CM by $\mathbb{Z}[\sqrt{-2}]$ while E_d has CM by $\mathbb{Z}[i]$, and no permutation of $(-2, 1, 1)$ satisfies $c_1 = 2c_2 = 2c_3$. Hence every classified curve inside X_d avoids the positive distinct full-magic transversal locus, and applying the imported Theorem 1.3 to $\Gamma = \Gamma_d^3$ of rank $3r$ gives finiteness with the stated C^{3r+1} shape. The bound is non-effective and yields no enumeration.

Squareclass passage. Writing $d = q^2k$, the isomorphism $E_k \rightarrow E_d$, $(X, Y) \mapsto (q^2X, q^3Y)$ commutes with doubling and scales x -coordinates, the difference and the interaction relation all by q^2 , preserving positivity and distinctness; every solution with difference in the squareclass $[k]$ descends to one of the finitely many on E_k . $\square$

6.2 The fibre $x_0 = 841$ is closed

Here $841 = x(2Q)$ is the centre of Robertson’s classical progression.

Proof of Theorem 6(ii). Fix $x_0 = 841$, write $x_1 = 841 + t$, $x_2 = 841 - t$, and put $f(x) = x(x^2 - 840^2)$. If both endpoints lie in $x(2E_{840}(\mathbb{Q}))$ then $f(841 + t)$ and $f(841 - t)$ are rational squares, so their product is one: $w^2 = f(841 + t)f(841 - t)$, which for $a = 841$ simplifies to the genus-two curve

$$C : w^2 = (1 - t^2)(841^2 - t^2)(1681^2 - t^2).$$

This is only a necessary condition, and that is enough: we close the superset. The even sextic makes C bielliptic. The quotient by $(t, w) \mapsto (-t, w)$ has rank 3, as independently corroborated by the PARI/GP record shipped in the companion repository; this contextual calculation is not used in the proof and does not close the fibre. The complementary quotient by $(t, w) \mapsto (-t, -w)$ is decisive: for $t \neq 0$ set $X = 1/t^2$, $Y = w/t^3$, then with $A = 841^2$, $B = 1681^2$, $L = AB$, $u = LX$, $v = LY$ the monic model is

$$E_- : v^2 = (u - A)(u - B)(u - L),$$

and $\text{rank } E_-(\mathbb{Q}) = 0$ with $E_-(\mathbb{Q})_{\text{tors}} \cong (\mathbb{Z}/2)^2$, the nonzero torsion sitting at $u \in \{A, B, L\}$. The rank and torsion are computed in proof mode and independently replayed on the official Magma Calculator V2.29-8 (RankBounds = (0, 0), proof flag true); the frozen transcript and machine-readable certificate ship in the companion repository, and Magma enters only as that hash-verified transcript. Every rational point of C with $t \neq 0$ maps to $E_-(\mathbb{Q})$; the three finite torsion points

give $t^2 = L/u \in \{1, A, B\}$, the point at infinity of E_- lifts to the two points with $t = 0$, and the two infinite points of the even model are irrational since its leading coefficient is -1 . Hence

$$t \in \{0, \pm 1, \pm 841, \pm 1681\}$$

for every affine rational point. Positive endpoint centres require $841 \pm t > 840$, i.e. $-1 < t < 1$, leaving only $t = 0$: the repeated three-entry boundary, not a distinct completion. The fibre contains no positive nonconstant three-transversal completion. $\square$

6.3 A fixed Bremner shadow is constant

Bremner's displayed degree-four solution [Bre01] decomposes into a rational constant part and a $\sqrt{3}$ -anti-invariant part. After dividing by the constant centre, the resulting rational line is

$$J + tB, \quad B = \begin{pmatrix} 88 & -153 & 65 \\ -23 & 0 & 23 \\ -65 & 153 & -88 \end{pmatrix},$$

where J is the all-ones matrix. The next proposition concerns only this one line; it is not a global obstruction to magic squares of squares.

Proposition 6 (Computer-assisted fixed Bremner shadow). *If every entry of $J + tB$ is a square in $\mathbb{Q}$, then $t = 0$.*

Proof. The four conditions $1 \pm 23t, 1 \pm 65t \in \mathbb{Q}^2$ imply

$$Y^2 = (1 - 23^2 t^2)(1 - 65^2 t^2).$$

For $t \neq 0$, putting $x = 1/t^2$ and $y = Y/t^3$ gives

$$E : y^2 = x(x - 529)(x - 4225).$$

Under the documented semantics of `ellrank`, the first two entries of the frozen PARI/GP output are unconditional lower and upper rank bounds; both are zero. The same record identifies the minimal model as Cremona curve 345345r4 [PARI, Cre]. For completeness, the torsion calculation is independently elementary. Reduction at 17 and 41 gives $\#E(\mathbb{F}_{17}) = 16$ and $\#E(\mathbb{F}_{41}) = 40$, so rational torsion has order dividing 8. All three nonzero rational 2-torsion points fail the standard halving criterion: for the roots 0, 529, 4225, the two required root differences are never both rational squares. Hence $E(\mathbb{Q})_{\text{tors}} = E[2](\mathbb{Q}) \cong (\mathbb{Z}/2)^2$.

The finite x -coordinates are therefore 0, 529, 4225. Since $x = 1/t^2$, a nonzero parameter would have to be $t = \pm 1/23$ or $t = \pm 1/65$. In each case one of the original four linear forms is 2, not a rational square. Thus no nonzero t is possible. $\square$

6.4 Limits of the separate finite classes

Theorem 7 (The finite-class stop). *Half classes $\lambda_n(P) = [Q] \in E_d(\mathbb{Q})/(2^n E_d(\mathbb{Q}) + E_d(\mathbb{Q})[2])$ are well defined for $P \in 2E_d(\mathbb{Q})$. On E_{840} , neither $P \mapsto x(P)$ nor the interaction of Theorem 5 factors through the finite class triple for any $n \geq 1$: the explicit pair $Q, Q + 2^n R$ has equal classes and unequal doubled x -coordinates. Hence no obstruction retaining only the three individual finite Kummer/2-Selmer classes can characterize the interaction. The certified true witness is the repeated triple (Q, Q, Q) ; a stronger statement whose hypotheses first impose the positive pairwise-distinct open locus is not excluded.*

Proof of Theorem 7. Well-definedness: if $2Q' = P = 2Q''$ then $Q' - Q'' \in E_d(\mathbb{Q})[2]$, so the class of a half modulo $2^n E_d(\mathbb{Q}) + E_d(\mathbb{Q})[2]$ does not depend on the half, and the doubling map exhibits the canonical isomorphism with $2E_d(\mathbb{Q})/2^{n+1}E_d(\mathbb{Q})$ (kernel exactly $2^n E_d(\mathbb{Q}) + E_d(\mathbb{Q})[2]$).

The countercertificate lives on E_{840} with the basis Q, R of Section 5. The two classical progressions have Kummer representatives $\delta(Q) = (7, 10, 70)$ and $\delta(R) = (14, 6, 21)$, reproducing the rank-two structure. Now put $H = Q + 2R$: then $\delta(H) = \delta(Q)$, yet

$$x(2H) = \frac{490152611881}{53187849} \neq 841 = x(2Q),$$

and $2H$ yields a genuinely distinct positive square progression (roots $667439/7293$, $700109/7293$, $731321/7293$). In general, for every $n \geq 1$ the points Q and $Q + 2^n R$ share their λ_n -class while their doubled x -coordinates differ — equality would force $2^{n+1}R = O$ or $4Q + 2^{n+1}R = O$ against free independence. Finally compare the half triples (Q, Q, Q) and $(Q, Q, Q + 2^n R)$: identical λ_n -class triples, yet the first doubled-centre triple satisfies the interaction $x_1 + x_2 = 2x_0$ and the second does not. Hence neither $P \mapsto x(P)$ nor the interaction factors through the finite class triples, for any n . $\square$

Remark 6 (Scope of the stop). The certified true witness is deliberately the repeated triple (Q, Q, Q) , so the theorem rules out exactly the obstructions that retain only the three individual finite Kummer/2-Selmer classes on the ambient doubled-point space. It does not exclude representative-level methods — covering curves, heights, valuations, a Mordell–Weil sieve — nor a separate statement whose hypotheses first impose positivity on the distinct-entry locus and pairwise distinctness. Sections 7 and 3.1 pursue precisely such finer invariants, along a different quantity.

7 The support law

Throughout, a primitive positive full-magic transversal configuration is written with the three square progressions (r_i^2, s_i^2, t_i^2) of common difference d , and

$$u_i = t_i - r_i, \quad v_i = t_i + r_i, \quad z_i = u_i/v_i \in (0, 1), \quad u_i v_i = 2d.$$

Theorem 8 (The support law). *For a positive pairwise-distinct full-magic transversal configuration put $u_i = t_i - r_i$, $v_i = t_i + r_i$, $z_i = u_i/v_i$. Then $u_i v_i = 2d$, $0 < z_i < 1$, and the interaction is exactly the five-term unit equation*

$$z_1 + z_1^{-1} + z_2 + z_2^{-1} = 2(z_0 + z_0^{-1}),$$

nondegenerate on the positive distinct-entry locus. For every finite prime set $\mathcal{S} \ni 2$, the number of root-primitive configurations with $\text{supp}(2d) \subseteq \mathcal{S}$, modulo common rational-square scaling, is at most

$$\exp(30^{15}(3|\mathcal{S}| + 1)).$$

This counting theorem by itself produces no enumeration; the independent fixed-support classification was proved in Section 3. At every odd prime the largest of $|v_p(z_i)|$ occurs at least twice; the admissible endpoint pairs obey the explicit $(-1/p), (2/p)$ table; and $\text{supp}(2d) \subseteq \{2, 3\}$ is impossible for a primitive configuration.

7.1 The torus identity and nondegeneracy

Substituting u_i, v_i into the interaction $s_1^2 + s_2^2 = 2s_0^2$ and using $4s_i^2 = u_i^2 + v_i^2$ gives, after division by $2d = u_i v_i$, exactly

$$z_1 + z_1^{-1} + z_2 + z_2^{-1} = 2(z_0 + z_0^{-1}) : \quad (\text{L.1})$$

multiplying by $z_0/2$, the five-term unit equation

$$\frac{z_0 z_1}{2} + \frac{z_0}{2z_1} + \frac{z_0 z_2}{2} + \frac{z_0}{2z_2} - z_0^2 = 1 \quad (\text{L.1}')$$

in the multiplicative group generated by the z_i .

Nondegeneracy. On the positive distinct-entry locus — all $z_i \in (0, 1)$, pairwise-distinct progressions — no nonempty proper subsum of the left side of (L.1') vanishes. Two exact facts drive every case. First, all z_i and z_i^{-1} lie in the squareclass of $2d$: indeed $z_i \cdot 2d = u_i^2$ and $2d/z_i = v_i^2$, so any relation of the shape $z_i = 2z_j$, $z_i z_j = 2$, $z_i = 2z_j^{-1}$ or $2s_i^2 = u_0^2$ would make 2 a rational square. Second, the interaction itself. A vanishing subsum containing none of the four positive terms is empty; one containing only positive terms is positive. The remaining cases pair $-z_0^2$ with one, two, or three positive terms. With *one*: $z_0 z_1/2 = z_0^2$ gives $z_1 = 2z_0$, hence $u_1^2 = 2u_0^2$ — excluded, and likewise for the other three singletons. With *three*: the subsum equals $1 - (\text{omitted term})$ by (L.1') itself, so the omitted term equals 1, giving $z_0 z_1 = 2$ or one of its three analogues — excluded the same way. With *two* same-index terms: $z_1 + z_1^{-1} = 2z_0$ gives $4s_1^2/(2d) = u_0^2/d$, i.e. $2s_1^2 = u_0^2$ — excluded. With *two* pure cross terms, $z_1 + z_2 = 2z_0$: combined with the interaction $(u_1^2 + v_1^2) + (u_2^2 + v_2^2) = 2(u_0^2 + v_0^2)$ and $v_i = 2d/u_i$, the two relations force, for $x = (u_0/u_1)^2$ and $y = (u_0/u_2)^2$, both $x + y = 2$ and $xy = 1$, hence $x = y = 1$ and $z_1 = z_2 = z_0$ — against pairwise distinctness (and symmetrically for $z_1^{-1} + z_2^{-1} = 2z_0$). With *two* mixed cross terms, $z_1 + z_2^{-1} = 2z_0$: the same elimination collapses to $(u_0^2 u_2^2 - 4d^2)^2 = 0$, i.e. $u_0 = v_2$ and $v_0 = u_2$, so $z_0 z_2 = 1$ — impossible with both in $(0, 1)$. Hence (L.1') is nondegenerate in the sense required by the subspace theorem.

7.2 The fixed-support count

Fix a finite prime set $\mathcal{S} \ni 2$ and restrict to configurations with $\text{supp}(2d) \subseteq \mathcal{S}$. Modulo common rational-square scaling of the entries, the triple (z_0, z_1, z_2) determines the configuration, and each z_i lies in the group of $\mathcal{S}$ -units of rank at most $3|\mathcal{S}|$ relevant to (L.1). Theorem 1.1 of Evertse, Schlickewei and Schmidt [ESS02] bounds the nondegenerate solutions of a five-term unit equation over a group of rank r by $\exp(30^{15}(r+1))$; applying it with $r \leq 3|\mathcal{S}|$ gives at most

$$\exp(30^{15}(3|\mathcal{S}|+1))$$

root-primitive configurations. The bound counts; it does **not** enumerate, and no effective procedure follows from it.

7.3 Tropical and Legendre laws

For odd p the six valuations occurring in (L.1) are $\pm\nu_i$ with $\nu_i = v_p(z_i)$; in a vanishing non-Archimedean sum the least valuation is attained at least twice, which proves that the largest $|\nu_i|$ occurs at least twice. At $p = 2$ the same argument applies to $\min\{\nu_1, -\nu_1, \nu_2, -\nu_2, 1 + \nu_0, 1 - \nu_0\}$. Writing $e = v_p(2d)$ and $a_i = v_p(u_i)$, the relation $u_i v_i = 2d$ gives $\nu_i = 2a_i - e$ with $0 \leq a_i \leq e$; call an index an *endpoint* if $a_i \in \{0, e\}$ and *interior* if $0 < a_i < e$. For odd p , an interior index puts p in every root of its progression: p divides both $u_i = t_i - r_i$ and $v_i = t_i + r_i$, hence t_i and r_i , and

then $4s_i^2 = u_i^2 + v_i^2$ forces $p \mid s_i$. Root primitivity therefore forbids three interior indices, so at least one index is an endpoint; and an endpoint attains the maximal value $|\nu_i| = e$, so the tropical tie forces a second endpoint: **at least two indices are endpoints at every odd support prime**. If exactly two indices are endpoints, comparing the unit squareclasses of the two least-valuation terms forces -1 to be a quadratic residue for the pair $\{1, 2\}$ and 2 for a pair containing 0 ; hence the exact endpoint pairs obey

$p \bmod 8$	allowed exact endpoint pairs
1	$\{0, 1\}, \{0, 2\}, \{1, 2\}$
3	none; all three indices are endpoints
5	$\{1, 2\}$ only
7	$\{0, 1\}, \{0, 2\}$ only

These are necessary local laws, not sufficient conditions.

7.4 Exclusion of support $\{2, 3\}$

By Lemma 1 of [PTZ15] every entry of a primitive configuration is $1 \bmod 24$, so all roots are coprime to 6 and $24 \mid d$. Assume $\text{supp}(2d) \subseteq \{2, 3\}$ and put $e_p = v_p(2d)$. At 2 , both u_i, v_i are even and exactly one has valuation 1 , so $v_2(z_i) = \pm(e_2 - 2)$; at 3 , exactly one of them is divisible by 3 , so $v_3(z_i) = \pm e_3$; there are no other primes. Hence $z_i \in \{2^{\pm(e_2-2)}3^{\pm e_3}\}$, and since $z \mapsto z^{-1}$ pairs the four sign choices, $f(z) = z + z^{-1}$ takes at most two values on the z_i . Three elements of a two-element set can satisfy $f(z_1) + f(z_2) = 2f(z_0)$ only if all three values agree, and f is strictly decreasing on $(0, 1)$, so $z_0 = z_1 = z_2$; with the same $K = 2d$ the formulas $u_i = \sqrt{Kz_i}$, $v_i = \sqrt{K/z_i}$ then force the three progressions to coincide, contradicting pairwise distinctness. This completes the proof of Theorem 8.

Remark 7 (Scope). Everything here is a necessary-condition layer on the common difference. The subspace-theorem count is imported [ESS02]; the congruence input is [PTZ15]; the tropical mechanism is elementary. No effectivity, no global support-size bound and no all-support emptiness follows, and none is claimed from this counting argument. The separate two-term S -unit route in Section 3 does give a finite enumeration for the specific complete support catalogues used there.

8 The four-prime frontier

Section 3 closes all supports contained in S_{19} . Here the extra primes are arbitrary; the remaining branches are not resolved by that finite-support result.

Let $\text{supp}(2d) = \{2, 3, p, q\}$ with p, q distinct primes different from $2, 3$, and put $U_i = (t_i - r_i)/2$, $V_i = (t_i + r_i)/2$, $N = d/2$, so $U_i V_i = N$ and $U_i^2 + V_i^2 = s_i^2$.

Proposition 7 (Endpoint incidence and edge factorization). *Each extra prime labels either no interior progression or one residue-permitted interior vertex, so at least one progression is endpoint at both p and q and its Pythagorean triple is primitive; the number of primitive progressions is $3 - \#(\{I_p, I_q\} \setminus \{\emptyset\})$. For every pair i, j , setting*

$$G = \gcd(U_i, U_j), \quad R = U_i/G, \quad S = U_j/G, \quad B = N/(GRS)$$

gives $U_i = GR$, $V_i = BS$, $U_j = GS$, $V_j = BR$ and the edge identity

$$s_i^2 - s_j^2 = (G^2 - B^2)(R^2 - S^2).$$

An interior-prime label splits its prime-power block on the two incident edges and leaves it whole on the opposite edge; the centre interaction becomes the equality $F_{10} = F_{02} = D \neq 0$ of the two canonical products of four difference factors. Two exact fixtures (area 210; the mixed area-30600 dropout) certify that neither two primitive progressions nor one primitive plus one interior progression is by itself contradictory — which is precisely why no exclusion is claimed at this level.

Proposition 8 (Balanced-dropout reduction). *If an extra prime is balanced at an interior vertex (its exponent split equally between the two legs), every other interior label is absent or co-located at the same vertex; every balanced branch has $d/4 = Ag_i^2$ with*

$$A \in \{6, 30, 60, 180, 84, 504, 1224\},$$

and primitive core $(3, 4, 5)$ or one of the six cores classified by Aebi [Aeb26]. If both extra primes are balanced at the central vertex, then $A = 6$ and, after scaling, a candidate forces a rational point on

$$w^2 = (1 - t^2)(625 - t^2)(2401 - t^2),$$

whose complementary bielliptic quotient (the method of Flynn and Wetherell [FW99]) is $y^2 = x(x - 111)(x - 93750)$, of rank zero with torsion $(\mathbb{Z}/2)^2$ — proof-mode computation replayed on the official Magma V2.29-8, entering as a hash-verified frozen transcript. The rational lifts give $t \in \{0, \pm 1, \pm 25, \pm 49\}$, positivity requires $|t| < 1$, and $t = 0$ repeats the progression: the central double-balanced branch is empty.

Proposition 9 (Outer core-area-6 exclusion and the addition curves). *If both extra primes are balanced at an outer vertex, the primitive core is $(3, 4, 5)$; with core entries 1, 25, 49 and central and opposite outer entries $25 + t$, $25 + 2t$, a candidate forces a rational point on*

$$W^2 = (t + 1)(t + 25)(t + 49)(2t + 1)(2t + 25)(2t + 49),$$

whose complete set of rational t -coordinates is

$$\{-49, -25, -49/2, -25/2, -1, -1/2, 0\}$$

(certified-complete point list, frozen transcript). Positivity of both new progressions requires $t > -1/2$, so only $t = 0$ survives and repeats the core: the outer double-balanced core-area-6 branch is empty on the positive distinct locus. For the residual central cores, normalizing the congruent-number curve as $E_n: y^2 = x^3 - n^2x$ with fixed central x -coordinate a , a secant slope z between P, Q with $x(P) + x(Q) = 2a$ lies on

$$H_{n,a}: Y^2 = (z^2 - (2a - n))(z^2 - 2a)(z^2 - (2a + n));$$

for the six Aebi central cores (areas 30, 60, 180, 84, 504, 1224) the Jacobian ranks are 2, 2, 3, 3, 4, 5, so ordinary genus-two Chabauty does not apply. In the first two cases, two-cover descent leaves a singleton fake two-Selmer set, identifying a unique covering class for elliptic or quadratic Chabauty in the sense of Bruin [Bru03] — a target, not a result: the rational points remain undetermined.

The exact open boundary

No arbitrary four-prime exclusion is claimed. What remains open at this frontier, stated exactly: the six Aebi central genus-two fibres (ranks 2, 2, 3, 3, 4, 5 above), the outer fixed cores other than the area-6 one, and the persistent unbalanced four-prime primitive cores. The continuation of the residual arithmetic growing out of the area-30/60 covering classes lives in the companion repository's dossiers with its own open gates, and no statement about it is imported here.

9 Open questions

The difference-side approach yields both constructions and complete exclusions on specified domains. It produces two infinite families of $(8, 7)$ arrays, classifies the $(9, 7)$ arrays supported on primes at most 19, and closes particular full-magic fibres. The remaining questions are arithmetic questions about the interaction, rather than about the seven-line form itself.

First, the four-prime reductions of Section 8 leave fixed core fibres and unbalanced branches beyond the S_{19} domain. Deciding these branches uniformly as the two additional primes vary would extend the finite-support results substantially. Second, fixed-squareclass finiteness is non-effective: an effective bound or a terminating enumeration for an arbitrary prescribed squareclass remains to be found. Third, the model of Section 5 separates the generic cone map from the rational-square condition needed at fixed difference. Its other maps, covering curves and arithmetic fibres remain possible tools; the S_5 result concerns the displayed map only. Likewise, the finite-class counterertificate leaves methods using representatives, heights and valuations available.

The companion repository records three further continuations: a residual cover over $\mathbb{Q}(\sqrt{218})$, an odd lift tower, and a marked class in a degree-90 S -class group. Their exact unresolved questions and supporting records are separated from the principal theorems in Appendix C. None decides the general magic-square problem. A complete two-term S -unit list gives a finite procedure for one fixed support, as Section 3 demonstrates; it does not by itself decide the problem over all supports.

A The bounded $(8, 7)$ census

A.1 The exact minimum census

The census definition is frozen: nine distinct positive integer entries; exactly eight positive squares; exactly seven of the eight ordinary lines with one common sum; quotient by the geometric D_4 action; height equal to the largest positive square root among the eight squares.

Proposition 10 (Exact minimum). *Under this definition there are 0 classes of height ≤ 46 and exactly 9 classes of height ≤ 47 , of which exactly two have a nonsquare centre.*

Certificate argument. Two independently implemented exact enumerators in the companion repository agree: `census.reference.py` scans integer progression centres and common differences, while the census family of `verify.py` builds full progressions from root triples and partial ones from root pairs. At height 46 there are twelve full three-square progressions, all with distinct common differences; two full progressions of one difference are necessary, so the census is empty. At height 47 the only difference supporting two full progressions is 840:

$$(1, 841, 1681) = (1^2, 29^2, 41^2), \quad (529, 1369, 2209) = (23^2, 37^2, 47^2),$$

There are five raw partial progressions satisfying the square-height bound. Two repeat entries when combined with the full progressions and therefore fail nine-way distinctness:

$$(841, 1681, 2521), \quad (1369, 2209, 3049).$$

The only compatible partial progressions are

$$(121, 961, 1801), \quad (169, 1009, 1849), \\ (256, 1096, 1936).$$

For each partial progression the six role assignments collapse to three D_4 classes, giving $3 \times 3 = 9$; the last two partials carry their nonsquare in the middle, and assigning either to the centre role gives one centre-nonsquare class each. The census is replayed from scratch by the companion verifier (`verify.py`, census family), and all nine canonical representatives are frozen in its expected certificate. $\square$

The three shifts $t = 9, 96, 144$ of the $n = 1$ carrier antidiagonal (265, 1105, 1945) produce exactly the three partial components of the minimum layer — stated carefully: only three designated orientations are literal antidiagonal retractions of the fixed source orientation; the other six classes are role reassignments. **The public census is frozen at the height-47 bound above**; no figure at any other bound is claimed here.

B Least-shift details and computational certificates

B.1 The complete case analysis for Proposition 3

Setting and notation as in Section 4: $n \geq 2$, $(y, z) \mapsto (y + 2z, y + 3z)$ from (11, 15), induction inequalities

$$z > y, \quad 2y > z, \quad 5y > 3z, \quad 5z > 6y, \quad 8y > 5z, \quad z \geq 15,$$

$t_0 = 6z^2$, $h = t_0 - t$ with $0 < h < 6z^2$, and at t_0 the three entries are $C - t_0 = Y^2$ ($Y = yz + 1$), $E - t_0 = H = 4z^4 - z^2 + 1$, $G - t_0 = X^2$ ($X = xz - 1$).

The C entry. If $C - t = (Y + a)^2$ then $h = a(2Y + a)$. If $a \geq 5$ then $h \geq 5(2Y + 5) = 10yz + 35 > 6z^2$ because $5y > 3z$; hence only $a \in \{1, 2, 3, 4\}$.

The G entry. If $G - t = (X + b)^2$ with $b \geq 2$ then $h \geq 4X + 4 = 4yz + 8z^2 > 6z^2$ (here $X = xz - 1$ and $x = y + 2z$ along the recurrence); thus $b = 1$ and necessarily $h = 2X + 1 = 2yz + 4z^2 - 1$.

The E entry. The exact neighbouring squares are

$$(2z^2 - 1)^2 < H < (2z^2)^2 < (2z^2 + 1)^2 < H + 6z^2 < (2z^2 + 2)^2,$$

so $E - t$ can be square only for $h = z^2 - 1$ or $h = 5z^2$.

Excluding CE. Equating $a(2Y + a)$ with the two possible E increments gives, for $a = 1, 2, 3, 4$ respectively:

a	equality with $z^2 - 1$	equality with $5z^2$
1	$z(z - 2y) = 4$	$z(5z - 2y) = 3$
2	$z(z - 4y) = 9$	$z(5z - 4y) = 8$
3	$z(z - 6y) = 16$	$z(5z - 6y) = 15$
4	$z(z - 8y) = 25$	$z(5z - 8y) = 24$

The first column is negative in every row because $2y > z$. In the second column, rows 1 and 2 equate positive multiples of $z \geq 15$ with 3 or 8; row 3 has $5z - 6y > 0$, and for $z > 15$ the product exceeds 15 while at $(y, z) = (11, 15)$ it equals 135; row 4 is negative because $8y > 5z$. Branch CE is impossible.

Excluding CG. Equating the four C increments with $2yz + 4z^2 - 1$ gives respectively

$$4z^2 = 4, \quad 2z(2z - y) = 9, \quad z(z - y) = 4, \quad 2z(2z - 3y) = 25.$$

The first and third contradict $z \geq 15$; the second and fourth equate an even integer with an odd one. Branch CG is impossible.

Excluding EG. The G increment exceeds $z^2 - 1$, their difference being $2yz + 3z^2 > 0$; equality with $5z^2$ would give $z(2y - z) = 1$, contradicting $z \geq 15$ and $2y > z$. Branch EG is impossible.

Every pair is excluded, so no positive pair-closing shift is smaller than $6z^2$; the shift $6z^2$ itself closes C and G , and Theorem 4(i) shows it is nondegenerate. The proof is uniform in n , uses no factorization of a particular Δ and no scan over t ; the packaged factor-pair census for $1 \leq n \leq 6$ is corroboration only. The statement deliberately starts at $n = 2$: at $n = 1$ the shift 9 is smaller than the uniform $6z^2 = 96$.

B.2 Certificate discipline

The companion repository distinguishes two verification layers. Its original Python verifier re-derives the elementary and symbolic witnesses (the integral line-sum Smith data; the recurrence identities and least shifts; the bridge divisor check and the (79, 65, 89) completion; the elliptic-orbit witnesses with denominator clearing; the saturation and monodromy data; the fibre algebra and pulled-back t -lists; the fixed Bremner-shadow quotient, finite-field counts and torsion-halving test; the half-class countercertificate; the torus, endpoint-table and three-block identities; the four-prime incidence and fixed-fibre identities) using exact integer and symbolic arithmetic. A deterministic high-precision decimal quadrature corroborates only the printed decimal of the exact Haar-measure density formula; it is not a logical input to the density proof. Computations performed using external computer algebra (the rank, torsion, complete point-list and Selmer computations attributed in the text to the official Magma Calculator V2.29-8, plus two PARI/GP quotient records) enter as *hash-pinned inputs and normalized output records*. These frozen transcripts and records are authenticated by the verifier, which checks their hashes and parsed assertions; it does not independently prove those external calculations and never executes either system. The fixed Bremner-shadow rank is supplied by a hash-pinned PARI/GP input/output pair; the verifier independently checks every elementary step surrounding that rank result. The exact file-level map is generated with the verifier and covered by the repository manifest.

C Supplementary support reductions and continuations

The first subsection gives an elementary counterpart to the Aebi-based exclusion. The later subsections distinguish necessary incidence data and computational organization from the unresolved Diophantine questions.

C.1 The elementary endpoint route

For comparison, we retain the local endpoint argument and valuation restrictions. In this subsection the configuration is primitive and satisfies the full-magic interaction.

Throughout, $\text{supp}(2d) = \{2, 3, p\}$ with $p \neq 3$ prime, and $d/2 = XYZ$ with $X = 2^\alpha$, $Y = 3^\beta$, $Z = p^\gamma$. The two technical stages are stated first; the headline Theorem 2 subsumes both by a different global argument, and its proof is independent of theirs.

Proposition 11 (All-endpoint case, $p \equiv 3 \pmod{8}$). *There is no primitive positive full-magic transversal configuration with $\text{supp}(2d) = \{2, 3, p\}$ and $p \equiv 3 \pmod{8}$.*

Proof. By the endpoint table of Section 7, $p \equiv 3 \pmod{8}$ forces all three indices to be endpoints at p , at 3 (same residue argument), and — by the opposite-parity mechanism at 2 — at the 2-power as well. Each progression's equal-area Pythagorean triangle is then primitive, and its legs are a unitary partition of the three whole blocks X, Y, Z . Apart from the impossible split $\{1, XYZ\}$,

only $\{X, YZ\}$, $\{Y, XZ\}$, $\{Z, XY\}$ remain, and three pairwise-distinct progressions must use all three. Factoring the hypotenuse equations of the last two gives

$$Y^2 - 1 = 2XZ, \quad Z^2 - 1 = 2XY,$$

whose difference factors as $(Y - Z)(Y + Z + 2X) = 0$. Positivity forces $Y = Z$, after which $Y(Y - 2X) = 1$, impossible for positive prime-power blocks. $\square$

Proposition 12 (One-interior law, $p \equiv 5, 7 \pmod{8}$). *Let $\text{supp}(2d) = \{2, 3, p\}$ with $p \equiv 5$ or $7 \pmod{8}$. The all-endpoint branch is impossible by the three-block argument, so $\gamma = 1$ is excluded, and for $\gamma \geq 2$ exactly one index is interior at p . If $p \equiv 5 \pmod{8}$ the centre progression is the interior one and exactly one of*

$$v_p(X^2 + Y^2) = 2h, \quad v_p(1 + Y^2) = 2h, \quad v_p(1 + X^2) = 2h$$

holds, with $h = v_p(s_0) \geq 1$; if $p \equiv 7 \pmod{8}$ one outer progression is interior and exactly one of the six ordered differences

$$v_p(2X^2 - Y^2), v_p(2Y^2 - X^2), v_p(2 - Y^2), v_p(2 - X^2), v_p(2Y^2 - 1), v_p(2X^2 - 1)$$

equals $2h$, with $h = \min(a, \gamma - a)$. These are necessary branch laws, not contradictions; their exact certificate covers all nine identities.

The nine residual branches of Proposition 12 need not be analyzed one by one: the headline theorem closes every case, including $p \equiv 1 \pmod{8}$, at once.

An elementary proof for primitive exact support $\{2, 3, p\}$. By the endpoint table, every index is an endpoint at 3 and at least two indices are endpoints at p . Those two indices give two *distinct primitive* Pythagorean triangles of the same area $d/4$, whose area has exactly three distinct prime factors. Writing $X = 2^\alpha = 2x$, the three endpoint types satisfy

$$YZ = x^2 - 1, \quad Y^2 - 1 = 4xZ, \quad Z^2 - 1 = 4xY.$$

Combining the first with each of the others reduces both mixed pairs to one equation

$$W(W^2 - 1) = 4x(x^2 - 1),$$

where W is the remaining odd block. The 2-adic valuation of the right side forces $W \equiv \pm 1 \pmod{2x}$, while monotonicity of $W(W^2 - 1)$ gives $x < W < 2x$; the only residue in that window is $W = 2x - 1$, which forces $x = 2$, $W = 3$, and then the third block equals 1 — not a prime power in the support. The remaining pair of endpoint types forces an equality of powers of two distinct primes, equally impossible. Hence no primitive positive full-magic transversal configuration has $\text{supp}(2d) = \{2, 3, p\}$ for any prime $p \neq 3$. $\square$

Remark 8 (Attribution and scope). The headline theorem is the squareclass consequence of Aebi's complete triangle classification proved in Section 3.1. The endpoint argument gives another proof of its primitive full-magic consequence, using only the stated endpoint restrictions. The uniqueness assertion fails already at four primes, as the difference-840 example shows. Section 3 nevertheless closes particular four-prime supports by a separate complete catalogue; arbitrary four-prime supports and global support-size control remain open.

C.2 The difference-side taxonomy

Consider a root-primitive positive full-magic transversal configuration. For each progression put

$$U_i = (t_i - r_i)/2, \quad V_i = (t_i + r_i)/2,$$

so $U_i V_i = d/2$ and $U_i^2 + V_i^2 = s_i^2$. Call its AP triangle *primitive* when $\gcd(U_i, V_i) = 1$, and let I be the set of vertices whose triangles are not primitive. A prime is interior at i precisely when it divides both U_i and V_i . By the support law in Section 7, neither 2 nor 3 is interior, and each other support prime is interior at at most one vertex. Thus

$$\#\{\text{primitive AP triangles}\} = 3 - |I| \geq \max\{0, 5 - \omega(2d)\}.$$

Indeed, each occupied vertex requires a distinct prime outside $\{2, 3\}$. The eight subsets of $\{0, 1, 2\}$ give eight labelled incidence cases, without asserting that any case is realized by a configuration.

The allowed interior vertices for an odd prime are $\{0, 1, 2\}$, $\emptyset$, $\{0\}$ and $\{1, 2\}$ for residues 1, 3, 5, 7 mod 8, respectively. An interior prime also requires $v_p(d) \geq 2$. For a fixed d , let n_j count the primes $p \notin \{2, 3\}$ with this exponent condition and $p \equiv j \pmod{8}$. In the relaxation that retains only these residue and exponent conditions, assigning distinct primes to all three vertices is possible exactly when

$$n_1 + n_5 \geq 1, \quad n_1 + n_7 \geq 2, \quad n_1 + n_5 + n_7 \geq 3.$$

To see this, apply Hall's matching criterion to the vertices and eligible primes. The displayed inequalities are its conditions for the central singleton, the pair of outer vertices, and all three vertices. The two outer vertices have identical neighbor sets, and every remaining subset condition follows from those three. This is an equivalence for the matching relaxation only; the Pythagorean and full-magic interaction equations still have to hold.

C.3 The outer-primitive reduction

A valuation record organizes possible branches but does not produce the required triangles. A further elementary reduction applies when the two outer triangles are both primitive. Set

$$G = \gcd(U_1, U_2), \quad R = U_1/G, \quad S = U_2/G.$$

Since $\gcd(R, S) = 1$ and $U_1 V_1 = U_2 V_2$, there is a positive integer B such that

$$(U_1, V_1) = (GR, BS), \quad (U_2, V_2) = (GS, BR).$$

The two primitive gcd conditions make G, B, R, S pairwise coprime. Exactly one factor is even, and it is divisible by four because the even leg of a primitive Pythagorean triangle is divisible by four.

Put $A = G^2 + B^2$ and $C = R^2 + S^2$. Exactly one of these norms is twice an odd integer; divide that norm by two to obtain odd integers A_0, C_0 . The full-magic interaction gives

$$2s_0^2 = s_1^2 + s_2^2 = (G^2 + B^2)(R^2 + S^2), \quad A_0 C_0 = s_0^2.$$

Hence A_0 and C_0 have the same unique positive squarefree kernel. Moreover, if an odd prime q divides both, pairwise coprimality permits division by B, S modulo q , giving

$$(G/B)^2 \equiv (R/S)^2 \equiv -1, \quad (s_1/(BS))^2 \equiv 2 \pmod{q}.$$

The last congruence uses $s_1^2 = G^2 R^2 + B^2 S^2$. Thus both -1 and 2 are quadratic residues modulo q , and $q \equiv 1 \pmod{8}$.

The common-kernel conclusion uses the central interaction as well as the two primitive outer triangles. Conversely, a common kernel does not supply the two outer-square equations. These necessary reductions leave the nondegenerate rational and integral point problems open.

C.4 Where the open objects live

The companion repository records three continuation boundaries in `dossiers/README.md`. These are status summaries of further arithmetic work. The shipped `scripts/verify.py` checks exact main-paper identities and the integrity and content of specified frozen computer-algebra outputs; it does not reproduce the deeper models and computations summarized in these dossiers. The mathematical conclusions depend on the proofs and the completeness statements of the imported computations, not on successful output checks alone.

- **Residual cover** (`dossiers/residual_cover/README.md`). The continuation from the area-30/60 covering classes of Section 8 records a residual cover q_R over $K = \mathbb{Q}(\sqrt{218})$. Neither a K -point nor emptiness is established. The recorded alternatives for the associated elliptic curve are rank two if a point exists, and rank one with a nontrivial divisible Tate–Shafarevich class if the cover is empty.
- **Odd lift** (`dossiers/odd_lift_n5/README.md`). The coefficient-3 result excludes nondegenerate rational specializations of the known generic coefficient-3 family satisfying both base reconstructions; it does not classify specialized Mordell–Weil points outside the generic subgroup. For coefficient 5, the reduction has two genus-two covers over $\mathbb{Q}(\sqrt{2})$. One is empty on the required real component; the surviving cover has a complete fake two-Selmer set of four classes, none eliminated. The associated degree-30 pair-field computation gives a rank-two lattice saturated at 2, 3, 5, 7, without a proved rank upper bound or finite index in the full Mordell–Weil group. A complete list of points with the required rational coordinate and a uniform result for larger odd coefficients remain open.
- **Degree-90 S -class** (`dossiers/degree90_sclass/README.md`). The marked ideal class $[A_R]$ in a degree-90 S -class group remains undecided. The dossier records a field construction and structural reductions, but bounded short-vector and elementary-unit searches produced no witness. Their failure does not prove that the class is nontrivial; the remaining decision requires a global witness or a rigorous obstruction.

The limits of earlier approaches are recorded in `dossiers/EXHAUSTED_ROUTES.md`. These continuations do not decide every branch as the prime support varies. For a fixed support, the two-term S -unit route of Section 3 gives a finite reduction; the uniform all-support problem remains open.

D Bibliographic comparisons

These notes record the scope of related constructions and restrictions. They complement the attributions attached directly to the results above.

D.1 A rationality filter

A fully magic square of nine distinct squares *does exist* over a number field of degree four. Bremner displayed the underlying magic square with eight entries square in $\mathbb{Q}(\sqrt{3})$ and centre 532, which is not a square in that field [Bre01]; adjoining $\sqrt{133}$ makes the centre $532 = (2\sqrt{133})^2$. Thus the degree-four completion is the elementary field extension of Bremner’s displayed object, not an additional construction attributed to him. The companion repository replays it as an exact verified witness. Bremner [Bre99], §3, separately constructs an example over a field of degree 27. The consequence deserves emphasis: there is no combinatorial or shape-only obstruction to the rational magic-square problem. Thus the obstruction to a rational solution must use arithmetic information that distinguishes $\mathbb{Q}$ from the displayed quartic field. This filter shapes the whole paper: the positive constructions of Section 4 show how close the rational story comes, and the support laws of Sections 3.1 and 7 are genuinely rational-arithmetic statements, invisible over a larger field.

D.2 The entry side: what is known about a hypothetical solution

A hypothetical rational solution is constrained through the prime factors of its *entries*; throughout this paragraph the square is taken in the primitive normalization (unit gcd of the nine entries). Rabern [Rab03] proved — Weisenberg’s restatement in [Wei23], §2.1, matches the primary on all six results — that every prime factor of an entry is odd; that a prime dividing the centre is $1 \pmod{4}$; that a prime $p \equiv 3, 5 \pmod{8}$ dividing a non-centre entry also divides the centre and the third entry of that line; that no $p \equiv 3 \pmod{8}$ divides any entry; that no $p \equiv 5 \pmod{8}$ divides a middle-side entry; and that a $p \equiv 3 \pmod{4}$ dividing a corner divides the two non-adjacent middle-side entries. Woll [Wol18] proved that a prime p dividing only the centre forces p to possess three consecutive nonzero quadratic residues, and counted the resulting residue classes; Labruna [Lab18] studied the order-three analogue over finite fields, where in characteristic greater than three a nontrivial magic square of squares attains exactly 3, 5, 7 or 9 distinct entries, each degree realized over infinitely many prime fields. Weisenberg [Wei23] added the order bookkeeping (a prime dividing exactly three entries attains its minimum order at least twice), the restriction $p \equiv 1 \pmod{24}$ for a prime dividing exactly two entries, consecutive-residue patterns for single-entry primes, a complete seven-arrangement list of the ways a prime factor can appear, and a proposal — not executed there — to organize searches by the Gaussian prime signature of the centre. A recent preprint of Hill [Hil26] announces a nonexistence proof; it is unrefereed, and nothing in this paper depends on it. Coumbe [Cou24] develops a Pell-transform approach and claims that the smallest entry cannot be unity or a prime square. We do not import that exclusion: the unrestricted difference-set statement used in its proof has the literal instance $120 + 120 = 240$, coming from the two primitive progressions $(7^2, 13^2, 17^2)$ and $(7^2, 17^2, 23^2)$. This does not refute a repaired distinct-summand statement or the final exclusion itself, but it prevents the published proof from being used here without repair.

All of these statements constrain *entries of a completed square*. The present paper works on a different quantity — the common difference of the three square progressions forced by the completion problem — and Section 7 depends on none of the entry-side results; the two layers meet only in the classical toolbox of quadratic residues and sums of two squares.

D.3 Provenance of the centre-true eight-square arrays

A related catalogued family must not be confused with the $(8, 7)$ metric: the *centre-true* arrays with a nonsquare centre g , eight distinct square entries around it, magic sum $3g$, and **six** magic

lines (three rows, middle column, both diagonals; the outer columns fail). Seven such arrays with centres

157441, 411625, 786361, 1199809, 2033641, 12138289, 19953649

appear in Brown’s note on automedian triangles and magic squares [Bro] which lists the six roots of the top and bottom rows for each centre. The square middle-pair completion gives the six-line arrangement below. We include it to distinguish this catalogued construction from our $(8, 7)$ metric, without treating the centres or their outer rows as new. For example, at $g = 157441$:

$$\begin{pmatrix} 421^2 & 49^2 & 541^2 \\ 481^2 & g & 289^2 \\ 149^2 & 559^2 & 371^2 \end{pmatrix},$$

whose six stated lines each sum to $3g = 472323$. Brown also states that a Parker centre root needs more than two distinct primes $1 \bmod 4$; that statement is non-refereed, its proof scope has not been established here, and it is not an input to any of our proofs.

D.4 Progressions in elliptic coordinate sets

Harrison–Mudgal–Schmidt [HMS26] give uniform rank-dependent bounds for progressions inside one elliptic coordinate set. This is complementary to the three-coordinate interaction studied here and does not decide its positive distinct-entry locus. The fixed-squareclass finiteness proof in Section 6 instead applies the explicit classification of elliptic curves in a linear x -coordinate relation due to Caro–García-Fritz [CGF25].

E Reproducibility

E.1 Engines

All theorem-level arithmetic in this paper is exact: integer and symbolic computation in Python with implementation-separated replays where a result is load-bearing, and elliptic-curve rank, torsion, and rational-point computations in the recorded software proof modes where available. Results attributed to the official Magma Calculator V2.29-8 (the rank-zero quotients of Sections 6 and 8, the certified-complete point lists, the descent transcripts) enter the record exclusively as **hash-verified frozen transcripts**: the verifier checks digests and parsed assertions and never executes that software. One accompanying PARI/GP record corroborates only the non-load-bearing rank-three quotient of Section 6; a second pinned record supplies the unconditional rank-zero interval for Proposition 6, while the verifier independently checks its quotient map, finite-field counts, torsion and terminal parameter elimination. No floating-point quantity feeds any conclusion. The checked-in manuscript was built deterministically with MiKTeX-pdfTeX 4.23 (MiKTeX 25.12), using the three-pass `pdflatex` route at the fixed `SOURCE_DATE_EPOCH`; the source also supports the Tectonic route documented in the companion repository. It is distributed as a public preprint companion artifact.

E.2 The smooth-support certificates

The complete total 3649 in Theorem 3 is imported from von Känel–Matschke [vKM16]; the corresponding witnesses and all downstream triangles, midpoint tests and $(9, 7)$ arrays are rebuilt locally. The producer and verifier do not import each other’s code: one enumerates $k + (m - k) = m$ and

tests $m + k$, while the other starts with $k + m = m + k$ and joins the other additive record. Their D_4 canonicalizations are also separate. These checks validate the finite deduction, not the source theorem’s original height bounds or sieve.

E.3 Data policy

Catalogued third-party data is cited and, where licensing is unresolved, **never bundled**: the Fitivalu tables are referenced by locator and digest [Fit] (URL, file name, byte count and SHA-256 in the bibliography entry and in the repository source lock), and third-party PDFs are cited by locator without redistribution. Web catalogue records (Boyer’s pages [Boy], Brown’s note [Bro]) are used under dated locators recorded in the repository’s source locks. Bounded censuses are labelled with their frozen bounds and are never used as evidence for an unbounded statement; where a census executable has a recorded interface defect, only its frozen default output is cited.

E.4 Manifest

The companion repository ships a manifest of SHA-256 digests covering every shipped source byte of the public package — scripts, certificates, transcripts, documentation and this manuscript’s sources; the compiled PDF (a deterministic function of the pinned sources on a fixed TeX distribution) and the regenerated results directory are deliberately unpinned. The original verifier has a frozen expected output whose digest is pinned inside the verifier. A separate exact verifier covers the smooth-support catalogue and its own frozen expected receipt. The original verifier compares its output with the frozen expectation byte for byte. The smooth-support verifier compares exact JSON values, types and structure, independently of whitespace and object-key order. Both require their expected receipt and fail on its absence or on a failed comparison.

E.5 AI assistance

The author used OpenAI Codex with substantial assistance in mathematical exploration, proposed arguments, code, exact computational checks, manuscript preparation and adversarial review. The author selected the research direction and is responsible for the submitted claims. Specific historical model versions are not reliably documented here and are not inferred from dates. AI-assisted reviews and computational replays are not independent specialist review or an end-to-end formal verification of the manuscript.

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